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Analyst Coverage Networks and Corporate Financial Policies

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Abstract. We use the setting of analyst coverage networks to shed light on the nature of peer effects in financial policies. First, we use the “friends-of-friends” approach and exploit the fact that analyst coverage networks partially overlap to identify endogenous peer effects, in which firms respond directly to the capital structure choices of their peers, separately from contextual effects, in which they respond to their peers’ characteristics. We further show evidence that analysts facilitate these peer effects through their role as informational intermediaries. Analyst network peer effects are distinct from industry peer effects and are more pronounced among peers connected by analysts that are more experienced and from more influential brokerage houses. Finally, the analyst peer effects become weaker after exogenous reductions in common coverage as a consequence of brokerage closures.

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Keywords: analyst network • friends of friends • peer effects • equity shock • capital structure

1. Introduction

A growing literature shows that a number of corporate policies, including security issuance, investment, payout policy, and stock splits, are influenced by a firm’s peers.¹ Although these results have been illuminating, several challenges remain. First, although the empirical approaches suggest the existence of some form of social interaction effect, separating the effect of peers’ policy choices from the effect of their characteristics remains challenging. Second, to the extent that firms do respond to the policy choices of their peers, the mechanisms through which firms learn about, and are influenced by, the choices of their peers are still not fully understood. In this paper, we use the setting of security analyst networks to shed light on both these issues.

As discussed by Manski (1993), a positive association between a firm’s policy and that of its peers can arise from multiple sources. First, peer firms may receive common shocks or share common unobserved characteristic that make them follow similar policies (what Manski calls “correlated effects”). Second, there are two types of potential social interaction effects, what Manski refers to as “endogenous effects” and “contextual effects.” Endogenous effects arise when firms respond directly to the financial policy choices of their peers. Alternatively, a firm’s choices may be influenced by the characteristics of

its peer group, giving rise to contextual effects. In a linear model, when the expected behavior of the group is a function of the average group characteristics, these effects cannot be separately identified, which Manski terms the reflection problem. Distinguishing between exogenous and endogenous effects is relevant because, for example, policy interventions such as targeted industry tax subsidies for debt financing, may influence the financial policies of peers while leaving their fundamentals unchanged. These policies may generate multiplier effects through endogenous peer effects (Glaeser et al. 2003) but are unlikely to if only contextual effects are present.

Previous approaches to identifying peer effects in corporate financial policies have used either random group assignment (Shue 2013) or exogenous characteristics of peer firms as instruments (Leary and Roberts 2014, Kaustia and Rantala 2015) to separate social effects from correlated effects. Yet, neither approach can fully break the reflection problem, because it is unclear whether firms are responding to their peers’ financial policy choices themselves or to the peer characteristics that drove those choices. For example, suppose a firm’s peer receives a shock to its investment opportunities and issues equity to fund it. If the focal firm issues equity in response, it is not clear whether they are responding to the peer’s

equity issuance or to the change in peer's investment opportunities that prompted it.²

We exploit the structure of analyst coverage networks (i.e., groups of firms covered by a common analyst) to better identify the presence of endogenous social effects in financing decisions, separately from contextual effects. Kaustia and Rantala (2015, 2021) argue that peer groups can be defined by the set of firms covered by the same sell-side equity analysts because these common coverage choices reflect economic linkages among firms. An additional advantage of this setting, however, is that unlike other peer group definitions such as industries, groups defined by common analyst coverage partially overlap. To distinguish endogenous peer effects from exogenous peer effects, we exploit the fact that we can observe intransitive triads in the analyst network. That is, we can observe firm triads i, j , and k , such that firms i and j and firms j and k have common analysts, whereas firms i and k do not have any. As shown by Bramoullé et al. (2009) and Goldsmith-Pinkham and Imbens (2013), this is a key property of the analyst coverage network that allows for identification of peer effects by exploiting the “friends of friends” approach. By using an exogenous characteristic of the peers of a firm's peers, we can construct an instrument that is correlated with peer firm's financial policies but is arguably uncorrelated with their characteristics, enabling identification of endogenous effects separately from contextual effects. Thus, our instrument is constructed using shocks to characteristics of firms that do not share a common analyst with the firm in question. This makes it hard to attribute the correlations we uncover to analysts choosing to cover firms with common unobserved characteristics.

A second advantage of the analyst network setting is that it enables us to study the role security analysts play in transmitting financial policy-relevant information across firms, highlighting a new mechanism facilitating peer effects in financial policies. The role of sell-side analysts in acquiring, analyzing, and disseminating information for investors has been much studied.³ In their role as information intermediaries, analysts may also facilitate the flow of information between firms. For example, Martens and Sextroh (2021) show that firms are more likely to cite each other's patents when they share a common analyst, suggesting that analysts facilitate inter-firm information transfers. Analysts not only learn about firm performance and prospects from company disclosures and conversations with managers, but also produce processed information in the form of research reports and recommendations. To the extent firm managers pay attention to analyst reports, they may convey information about the analyst's assessment of market conditions and preferred firm policies, and so on. Managers may pay attention to analyst reports either because they believe it to contain value relevant information or in their effort to cater to analyst preferences (Degeorge et al.

2013). Because analysts typically cover a portfolio of firms, the two-way communication between an analyst and management can also result in the propagation of financial policies across the firms in an analyst's portfolio. We show evidence consistent with this mechanism.⁴

We focus on financial policies such as leverage, debt issuance, and equity issuance. We classify all firms that share a common analyst with a firm as its “analyst peers” and relate the firm's financial policy to the weighted average financial policy of its analyst peers. We use the number of common analysts between the firm and its peer firms as the weights. This methodology gives rise to a network, which we refer to as the analyst coverage network, that is, the graph where the firms are the nodes and the weighted edges between two firms are the number of common analysts between the firms.

We begin by documenting a positive association between a firm's financial policy and that of its analyst peers. The association holds for leverage in levels and changes and debt and equity issuance decisions and is robust to controlling for firm characteristics, analyst peer characteristics, the average policies of industry peers, and industry peer characteristics. When we differentiate between within industry analyst peers and outside industry analyst peers, we find that the association extends to both sets of peers.

Firms in the same analyst network may pursue similar financial policies simply because they share similar characteristics or face a similar environment. This is especially likely given that analysts cover firms with common underlying economic features. Firms with common analysts may also follow similar policies due to the preferences of their common analysts (Degeorge et al. 2013). We control for such correlated effects following the procedure of Leary and Roberts (2014). Specifically, we use idiosyncratic equity return shocks as an exogenous source of variation in peer firm financial policy (and possibly characteristics) and relate it to a firm's financial policy. A large prior literature in finance shows that firms change their leverage and equity issuance decision in response to changes to their stock price (Marsh 1982, Baker and Wurgler 2002), which supports the relevance of our instrument. To the extent we are able to isolate idiosyncratic shocks to peer firms' equity values, the shocks are unlikely to be correlated with the characteristics of the focal firm, and thus any peer effects we document are unlikely to include correlated effects. In constructing the idiosyncratic equity shock, we estimate an augmented market model controlling for the average returns of analyst peers.

We estimate reduced-form regressions that establish a significant association between a firm's financial policy and the idiosyncratic return shocks of its analyst network peers. This association is robust to controlling for the financial policies, characteristics, and return shocks of the firm's industry peers, suggesting that the analyst network effect is distinct from an industry effect.⁵ These

results apply to leverage, net and gross equity issuance, and the debt-equity choice. We also show that these results extend to seasoned equity offerings (SEOs) in particular: Higher idiosyncratic returns of analyst coverage network peers increase the probability that the focal firm issues new seasoned equity in the next year. We further find that SEOs that follow those of analyst network peers are associated with higher announcement returns, consistent with previous evidence that analysts reduce information asymmetry around security issuances and of information spillovers across firms in the analyst's coverage network (Chan and Chan 2014). We also use the idiosyncratic shock to peer firms' stock prices as an instrument for peer firm financial policies in a two-stage least-squares (2SLS) specification and document a positive association between a firm's financial policy and analyst peer policies that is distinct from correlated effects, suggesting the existence of at least one form of social effects among analyst networks.

To separate endogenous effects from contextual effects, we define a firm's indirect peers as those firms that share a common analyst with other firms in the focal firm's analyst network but do not have an analyst in common with the focal firm itself (i.e., "friends of friends"). To illustrate, suppose we observe firms i , j , and k , such that firms i and j and firms j and k have common analysts, whereas firms i and k do not have any. We then use the idiosyncratic equity shock of firm k as an instrument for the financial policy of firm j to document its influence on i 's financial policy. The exclusion restriction for this approach is that the indirect peer's (firm k 's) return shock should influence firm i 's financial policy only through its influence on the direct peer's (firm j 's) financial policy and not otherwise. One necessary condition is that firm i does not respond directly to its indirect peers' financial policies. This is reasonable given that the set of firms two analyst connections away is large and heterogeneous, consisting of firms that not only have no common analysts but are also mostly in different industries.⁶

Identification also requires that firm k 's equity shock should not be correlated with firm j 's and firm i 's characteristics. To the extent we are able to isolate shocks to equity values that are idiosyncratic to firm k , it is reasonable to assume that it is uncorrelated with firm j 's characteristics. To the extent firm k and firm i do not even have common analysts, it is even more reasonable to expect idiosyncratic shocks to firm k 's equity to be uncorrelated with firm i 's characteristics. One potential threat to the former is if firm j is a supplier or customer to firm k , in which case economic shocks may have spillover effects. However, a major advantage of using shocks to firms outside of firm i 's analyst network as instruments is that we can control for the stock returns of the firms within the analyst network. In that case, any identification threat would have to come from a shock to firm j that is (a) significant enough to solicit a response from firm i but

(b) not reflected in firm j 's stock return, a scenario we consider unlikely.

We find robust evidence for endogenous peer effects using the friends of friend approach. The effects we document are economically significant. A 1-standard-deviation increase in peer firm average leverage is associated with a 0.482-standard-deviation increase in a firm's leverage. Peer effects are also present in a firm's decision to issue equity. A 1-standard-deviation increase in gross equity issuance likelihood by peers leads to a 0.520-standard-deviation increase in net gross equity issuance likelihood. Overall, after controlling for the endogeneity in the network formation, we find that peer firms in the same analyst coverage network affect each other.

Having established the presence of endogenous peer effects in analyst coverage networks, we next explore the potential mechanisms behind this behavior. Firms may respond to the financial policy choices of those with common analysts for several reasons. First, as argued by Kaustia and Rantala (2015, 2021), firms within the analyst network share economic linkages and face similar market frictions and investment opportunities. Thus, they may look to each other to learn about optimal financial policy. In addition, the analyst may play a role in the propagation of information about financial policy. Apart from regularly communicating with the firms, analysts also use common models to value the firms and benchmark them with one another. During the course of their communication and valuation, analysts obtain information that other firms may find valuable. Such information can be about the state of financial markets, growth opportunities, a particular financial policy, and so on, and may originate either from a particular portfolio firm or from the analyst. If learning about this information from analyst reports the firm managers choose to act on that information, then we expect financial policies to propagate from one firm to another within the analyst's portfolio.⁷ We follow a two-pronged strategy to show evidence consistent with this mechanism.

We first do a battery of tests to ensure that our results are specific to analyst networks and not solely a result of common industry linkages. First, as noted earlier, in all our tests, we control for industry average policies, either directly or through industry average return shocks. Second, we find similar results when we focus on the firms in the analyst network that are not from the same industry as the firm in question. Third, we estimate a placebo test in which we define pseudo peer groups as firms in the same industry as a firm's direct analyst peers but that do not have a common analyst with the firm in question. We find no evidence of peer effects in this sample. These results are robust across various industry definitions such as Hoberg-Phillips industry peers and Global Industry Classification Standard (GICS) code. Fourth, we exploit a quasi-natural experiment to shock the network structure. Specifically, we explore the closure and mergers

of brokerage houses as an exogenous reduction in common analyst coverage (Hong and Kacperczyk 2010, Kelly and Ljungqvist 2012, Israelsen 2016, Martens and Sextroh 2021). After identifying pairs of firms that suffered an exogenous reduction in common analysts, we find that the relation between the focal firm's leverage and equity issuance decisions and those of its peers become weaker after the shock. These results suggest that analysts play an important role in propagating information from one firm to another and the analyst network peer effect is not entirely driven by economic linkages that we do not observe.

Finally, we document cross-sectional variation in our estimated social effects that are unique to information transmission through the analyst network by testing whether more influential analysts are more effective at transmitting information across firms. Consistent with this idea, we find stronger peer effects among firms connected by analysts from brokerage houses with more "all-star" rated analysts and by more experienced analysts.

We make a number of important contributions. First, we add to the growing literature on peer effects in corporate policies. Although previous studies have shown evidence of social interaction effects in capital structure (Leary and Roberts 2014), dividends (Grennan 2019), and investment decisions (Fracassi 2017), the mechanisms and economic forces behind these effects are less well understood. We make two contributions relative to these earlier studies. First, we use the friends of friends approach to separately identify firms' responses to their peers' policy choices from the influence of peer characteristics, a distinction that was difficult to make with previous approaches. This approach can be productively used to document endogenous peer effects in other networks that partially overlap such as board networks and supply chain networks. Second, we show evidence suggesting that analysts may play an important role in propagating financial policies across firms. This information transmission represents an important underexplored channel through which peer effects may arise. We also contribute to the literature on the effects and implications of shared analyst coverage networks. Our results highlight an underappreciated role of security analysts in facilitating information flow across firms (not just between firms and investors). Although other studies have shown that common analyst coverage is associated with return comovement (Muslu et al. 2014, Israelsen 2016), we demonstrate that it can also lead to commonality in financial policies.

The rest of the paper is organized as follows. Section 2 discusses the related literature. Section 3 discusses our data and empirical methodology. Section 4 discusses the empirical evidence, and Section 5 concludes.

2. Related Literature

Our paper's main contribution is to show that analysts play an important role in facilitating peer effects in

capital structure policies and that analyst networks influence the way firms interact with one another. As such, our study is related to two main streams of literature. The first is the literature exploring the role of peer effects, or the interaction among agents, in corporate finance. Most of this literature provides evidence either for information transmission through networks of firms and/or managers or of the resultant correlated behavior among members of the network. Evidence of network effects in corporate policies is shown by Shue (2013), Leary and Roberts (2014), Kaustia and Rantala (2015), Fracassi (2017), and Grennan (2019), whereas Matvos and Ostrovsky (2010) and Pool et al. (2015) document peer effects among mutual fund managers.

Our paper is closest in spirit to Leary and Roberts (2014), which shows evidence that firms are influenced by their peers when making capital structure decisions. Although instructive, Leary and Roberts (2014) note several unresolved issues from their analysis, which they leave for future research. First, their empirical approach is not fully able to distinguish the nature of the social interaction effects among firms. That is, do firms respond directly to the financial policy choices of their peers, or do they respond to the change in peers' characteristics (e.g., investment opportunities, etc.) that prompted the financial policy choice? Second, they call for future work to better understand the mechanisms driving interdependence in financial policies. That is, how does information that informs financial policy choices get transmitted across firms?

We address both these issues by studying peer effects in financing decisions within analyst coverage networks. We propose analyst-firm interactions as one mechanism through which relevant information could transmit from one firm to another, and we further use the partially overlapping nature of analyst coverage networks to differentiate endogenous peer effects from exogenous or contextual effects. Our main model is an extended version of the Manski-type linear-in-means model similar to those studied in Goldsmith-Pinkham and Imbens (2013) and Bramoullé et al. (2009) (see also the survey by Blume et al. 2010).⁸

At the same time, we build on the large literature that studies the role of sell-side analysts in financial markets. Focusing on the traditional role of analysts as information intermediaries between firms and outside investors, some studies have shown evidence that analysts produce value-relevant information about the firms they cover (Womack 1996, Barber et al. 2001, Jegadeesh et al. 2004). Although we are not aware of any prior studies that show direct evidence of analysts recommending policy changes based on actions of other firms in their network, the combination of previous results suggest this is a plausible channel. First, Hilary and Shen (2013) show that analysts with more experience covering a given firm are better able to use information from that

firm's management forecast to improve forecast accuracy for other firms they cover. This suggests that some of the information analysts generate in covering one firm, such as industry-specific information, is relevant for understanding and valuing other firms in their network. This information transfer affects not only analyst forecasts but also investor behavior (Piotroski and Roulstone 2004, Muslu et al. 2014, Brochet et al. 2018).

Other studies suggest that, in addition to communicating information to investors, analysts also influence outcomes at the firms they cover. This can happen through analysts' monitoring role (Chen et al. 2015), catering to analysts' preferred policies (Degeorge et al. 2013), or learning from analyst feedback (Bae et al. 2022). Thus, the existing literature provides evidence that (i) analysts produce information that is relevant for firms across their coverage network, and (ii) communication runs both from the firm to the analyst and from the analyst back to the firm. This combination suggests that analysts may serve a role in facilitating information flow across the firms they cover. Some recent work supports this role. For example, Martens and SEXTROH (2021) show that pairs of firms are more likely to cite each other's patents if they have common analysts, which they interpret as evidence that analysts facilitate the flow of relevant competitive information between firms. Brown et al. (2019) survey corporate investor relations officers on what factors are important for managing their company's narrative. A majority of respondents agree that "knowledge about industry trends and/or competitors" is a very important service provided by sell-side analysts. Thus, there is some precedent for the idea that analysts facilitate information transfer across firms. The one element of our proposed channel that does not have direct precedent in the literature is whether analysts are influenced by the financial policy choices made by firms they cover or, at a minimum, communicate these choices to other network firms. Whether this channel operates is ultimately an empirical question to which we turn next. Our results provide additional evidence supporting information transfer across firms covered by a common analyst, and further show that this information transfer plays a role in propagating financial policies across connected firms.

3. Data and Empirical Methodology

3.1. Sample Selection

We obtain our data from standard sources: financial information from Compustat, stock price information from the Center for Research in Security Prices (CRSP), and analyst coverage information from the Institutional Brokers' Estimate System (IBES). From the overall CRSP-Compustat merged data set, we exclude financial firms (Standard Industrial Classification (SIC) codes between 6000 and 6999), utilities (SIC codes between 4900 and 4949) and government companies (SIC codes greater

than or equal to 9000). We then match the CRSP-Compustat sample to IBES and identify all firms that are connected to at least one other firm in the sample through a common analyst. We identify an analyst as following a firm in a fiscal year if that analyst makes at least one earnings forecast during the year, and the forecast is made at most six months before and three months after the end of the fiscal period. We also require the analyst to follow the pair of firms for at least two years in the entire sample for us to consider them to be connected through the analyst. Our final sample spans the period 1993–2015 and includes 41,411 firm-year observations.⁹

3.2. Empirical Methodology

We begin by documenting the extent to which financial policies of analyst peers are associated with a firm's financial policy. We do that by estimating the following regression:

$$y_{ijt} = \alpha + \beta_1 y_{-it}^{ACN} + \beta_2 y_{-ijt}^{IND} + \gamma_1' X_{-it-1}^{ACN} + \gamma_2' X_{-ijt-1}^{IND} + \gamma_3' X_{ijt-1} + \delta' u_i + \phi' v_t + \epsilon_{ijt}, \quad (1)$$

where the indices i , j , and t refer to firm, industry, and year, respectively. The dependent variables that we model are *Leverage*, Δ *Leverage*, *Net Debt*, *Net Equity*, and *Gross Equity*. All variables we use in our analysis are defined in Appendix A. *Net Debt*, *Net Equity*, and *Gross Equity* are dummy variables that identify debt and equity issuances. These take a value one if the firm issues debt (equity) in excess of 1% of total assets and zero otherwise¹⁰; X_{ijt-1} is the set of firm-specific controls. Following Leary and Roberts (2014), we include lagged (one period) values of *Log(Sales)*, *Market to book*, *Tangibility*, *Profitability*, and *Log(1 + Coverage)*. We also include the current size of the firm's network, *Log(1 + Connections)*, as our controls. *Coverage* is defined as the number of analysts following a firm in a given year. *Connections* is the size of a firm's network (degree: number of firms connected to firm i by common analysts); y_{-it}^{ACN} represents the weighted average value of the outcome variable for all the firms that are connected to firm i through common analysts in year t . The weights for each firm l in the analyst network equals the number of common analysts between firm l and firm i . Specifically,

$$y_{-it}^{ACN} = \frac{\sum_{i \neq l}^l n_{ilt} y_{lt}}{\sum_{i \neq l}^l n_{ilt}}, \quad (2)$$

where n_{ilt} represents the number of common analysts between firm i and firm l . In calculating y_{-it}^{ACN} , we use the financial policies of peer firms in the current year along with the current network structure. We use a weighted average instead of a simple average to give more weight to peer firms with more common analysts. Such peers may have a stronger influence on a firm's financial policy because there is a greater likelihood that one or more

analysts will transmit information across the firms. Our coefficient of interest is β_1 . We also include a set of weighted average peer firm characteristics (X_{-it-1}^{ACN}) as controls. These are the same set of characteristics as included in X_{ijt-1} and discussed previously, but we include the contemporaneous size of firm's analyst network ($\text{Log}(1 + \text{Connections})_i$). To calculate X_{-it-1}^{ACN} , we use the current network structure and lagged peer firm characteristics.

To distinguish the effect of analyst network peers from that of industry peers (Leary and Roberts 2014), we also control for the average value of the outcome variable for all other firms in the same industry (based on three-digit SIC code), y_{-ijt}^{IND} (excluding the firm i), and their average characteristics, X_{-ijt-1}^{IND} , as additional controls.¹¹ In all the regressions, except for those with changes in *Leverage* as the outcome variable, we include firm and year fixed effects. For the regressions with change in leverage as the outcome variable, we include industry and year fixed effects. We also include the independent variables in first difference form in this specification, except for the equity shock, analyst coverage and firm's network. The standard errors we estimate are robust to heteroskedasticity and clustered at the firm level.

As shown in Manski (1993), a significant β_1 can arise from one of three sources. First, it can reflect the fact that there are some unobserved similarities among firms in the same analyst network (correlated effects). These similarities may result in the firms choosing similar financial policies. Alternatively, it can arise from firms responding to either the behavior (endogenous peer effects) or characteristics (exogenous peer effects) of the peer firms. To control for correlated effects, following Leary and Roberts (2014), we use idiosyncratic shocks to the value of the peer firm's equity as an instrument for their financial policy (or characteristic). We define expected returns based on a one-factor market model augmented to include the excess return on the analyst network portfolio. We use the equally weighted portfolio returns of all firms that share a common analyst with a firm to calculate the excess returns. Although the excess return on the analyst network firms does not necessarily represent a priced risk factor, we include it to absorb any common shocks that may affect firms in the same analyst network.¹² For example, Muslu et al. (2014) and Israelsen (2016) show that shared coverage explains comovement and excess comovement between pairs of stocks with common analysts. Thus, we model the firm's stock return as

$$r_{it} = \alpha_{it} + \beta_{it}^M(r_{mt} - r_{ft}) + \beta_{it}^{ACN}(\bar{r}_{-it}^{ACN} - r_{ft}) + \eta_{it},$$

where the subscript t refers to time in months, r_{mt} and r_{ft} are the monthly return on the market and risk free asset, respectively, and $\bar{r}_{-it}^{ACN}$ is the equally weighted average return of all firms in the analyst network of firm i . We estimate this regression individually for each firm-year

using a five-year rolling window.¹³ We then calculate *Equity shock* for firm i in year t as the difference between the return on the firm's stock in year t and the predicted return based on the market and peer portfolio excess returns during the year and the loadings estimated using the data from the prior five years. We require firms to have at least 24 months of historical data to estimate the previous model. *Equity shock* represents the idiosyncratic shock to a firm's stock return. We then calculate the weighted average equity shock for the analyst network, $Equity\ shock_{-it}^{ACN}$, using the number of common analysts as the weights and the industry average equity shock, $Equity\ shock_{-ijt}^{IND}$, as the simple average equity shock for all firms in the same industry as firm i .

We use $Equity\ shock_{-it}^{ACN}$ as an instrument for y_{-it}^{ACN} and use a reduced form model and 2SLS to estimate its effect on firm i 's financial policy after controlling for industry corporate policy and industry characteristics. To the extent that $Equity\ shock_{-it}^{ACN}$ captures idiosyncratic shocks to the stock price and consequently leverage of analyst peer firms, it is unlikely to be correlated with firm i 's characteristics. To this extent, the reduced-form model and the 2SLS will isolate the social effects and exclude correlated effects. The specific identifying assumptions that we make for this are the following. First, for instrument relevance, we assume that $Equity\ shock_{-it}^{ACN}$ is correlated with the peer firm's financial policy either directly or indirectly through one or more characteristic. A large prior literature documents the important effect stock prices can have on firm financial policies (Marsh 1982 and Baker and Wurgler 2002, among others), and stock price changes often reflect changes in firm characteristics such as investment opportunities, expected profitability or risk, which in turn have been shown to be important determinants of firm financial policies. This suggests the relevance assumption will be satisfied in our setting, which is further supported by strong first-stage results below. The second assumption we make to isolate social effects is that $Equity\ shock_{-it}^{ACN}$ is uncorrelated with firm i 's characteristics. To the extent our procedure for defining $Equity\ shock_{-it}^{ACN}$ isolates truly idiosyncratic shocks, this assumption is likely to be valid.¹⁴ Leary and Roberts (2014) explore the properties of this instrument in depth and demonstrate its suitability in this context.

Our tests using $Equity\ shock_{-it}^{ACN}$ as an instrument will not be able to isolate endogenous peer effects from exogenous peer effects because the idiosyncratic shock to equity values can change, or reflect changes in, some (possibly unobserved) peer firm characteristic and firms may respond to the changes to peer firm characteristic as opposed to the changes in peer firm behavior.

To help isolate the endogenous peer effects from exogenous peer effects, we exploit the fact that analyst networks partially overlap with each other, and we can find intransitive triads in the analyst network. Bramoullé et al. (2009) show that one can identify peer effects when there

are intransitive triads, even when the analyst network is endogenous, say because analysts choose to cover firms with common (unobserved) features. In other words, in our network there are triads i, j , and k , such that firms i and j and firms j and k have common analysts, whereas firms i and k do not have any common analyst. Following the friends-of-friends approach outlined in Bramoullé et al. (2009), we use the characteristic of firm k (namely *Equity shock*) as an instrument for the financial policy of firm j to identify its influence on firm i 's financial policy. In our subsequent discussion we refer to firm k as an indirect peer of firm i . We use a slightly modified and in some senses a stricter version of the friends-of-friends approach proposed by Bramoullé et al. (2009). To identify endogenous peer effects, they only require that some of the indirect peers not be direct peers of the firm in question. If that is true, then one can use the characteristics of *all* the indirect peers as instruments for peer firm behavior. In our tests, we use the *Equity shock* of only the indirect peers that are not direct peers of the firm in question to instrument for peer firm behavior. By construction, there are no analysts in common between firms i and its indirect peers. Therefore, the specific instrument we employ is the simple average *Equity shock* of the indirect peers.¹⁵

The identifying assumptions necessary for us to isolate the endogenous peer effects are as follows. First, we require that the *Equity shock* of firm k be correlated with the behavior of firm j . This will happen as long as there are some social effects in analyst networks, which our earlier results confirm. Our second assumption has two parts to it. First, we require firm k 's equity shock to not be correlated with firm j 's characteristic. This is exactly the same as the exclusion restriction in our instrumental variable (IV) estimate and is also similar to the assumption in Leary and Roberts (2014). The second part of our assumption is that firm k 's equity shock not be correlated with firm i 's characteristic. This assumption is more easily satisfied than the previous assumption as, in addition to our instrument being idiosyncratic, firm i and firm k do not have any common analysts and they are often not even from the same industry.

Additionally, because firm k is in a different analyst network than firm i , we can control for the average equity return of firm i 's analyst network in this specification, which further rules out any confounding influence from common shocks to fundamentals that are not captured by the asset pricing model or spillovers from firm k to firm j and allows for identification even when the analyst network is endogenous. For example, if firm j is a supplier to firm k then a shock to firm k may impact firm j 's characteristics. This raises the possibility that we could still be picking up a contextual effect (i.e., firm i responding to firm j 's characteristic) in the friends-of-friends approach. However, because we can partial out the return shocks of the direct peers, such a spillover would have to be significant enough to impact firm i 's financial

policy, yet not be captured by firm j 's stock return. Finally, if firm i responds directly to firm k 's financial policy or characteristics, then we may not capture peer effects operating through analyst networks with this approach. However, we argue this is unlikely, given that the indirect peers consist of firms that do not share any common analysts with firm i and (in some specifications) are also in different industries.¹⁶ Thus, any social interaction effects should be much stronger among direct peers than between indirect peers. Furthermore, we perform cross-sectional tests later to demonstrate the relevance of analyst connections in transmitting policy choices across firms.

3.3. Summary Statistics

Table 1 provides descriptive statistics for the analyst networks. On average, a firm is connected to 42 other firms through common analysts. Interestingly, only 10.58 (29%) of these connections are from the same three-digit SIC code industry. The low percentage of within industry connections helps us independently estimate peer effects arising from both industry and analyst networks. Even with alternate definitions of industry (Fama-French and GICS codes), we find that the percentage of within industry analyst peers is uniformly less than 50%. We find that on average, two connected firms in our sample have 1.91 analysts in common. Surprisingly, this number does not vary much in the sample. The 25th percentile of the average number of common analysts by firm-year is 1.11, whereas the 75th percentile is 2.38. We find that firms within an industry are likely to have more common analysts as compared with firms across industries. Two firms within the same industry have on average 3.13 common analysts, whereas this number is only 1.56 for two firms from different industries.

We exclude from our analysis firms that are not connected to any other firm through common analysts. The variable *Connected Firms* identifies the percentage of firms that are connected to at least one other firm each year in the overall CRSP-Compustat-IBES sample. We find that about 94% of the firms in the overall sample are connected to at least one other firm. Thus, the unconnected firms, which we exclude, constitute only 6% of the CRSP-Compustat-IBES merged sample.

The average (median) number of indirect connections, defined as the pairs i and k , such that firms i and j and firms j and k have common analysts, whereas firms i and k do not have any, are 414.72 (382), and the 25th percentile of the number of indirect connections is 222, whereas the 75th percentile is 576. Most of the indirect connections are to firms in different three-digit SIC code industries. The mean (median) number of across industry indirect connections is 394.18 (359).

Table 2 reports the average value of the outcome variables we use in our analysis. We find that the average *Leverage* (change in *Leverage*) for the firms in our sample is 22% (1%). When we identify debt issuances as instances

Table 1. Analyst Coverage Network Summary Statistics

Direct peers	N	Number of connections					Connections (%)					Average number of analysts in common				
		Mean	SD	P25	P50	P75	Mean	SD	P25	P50	P75	Mean	SD	P25	P50	P75
Overall	41,411	42.09	27.33	21	38	58	0.29	0.3	0.04	0.16	0.46	1.91	1.07	1.11	1.52	2.38
Within industry (3 Digit-SIC code)	41,411	10.58	12.74	2	6	15	0.29	0.3	0.04	0.16	0.46	3.13	2.77	1.19	2	4
Across industries (3 Digit-SIC code)	41,411	31.5	26.04	11	26	46	0.71	0.3	0.54	0.84	0.96	1.56	0.73	1	1.27	1.8
Within industry (F-F Industry)	41,411	17.06	16	4	13	26	0.42	0.31	0.15	0.38	0.68	2.53	1.88	1.14	1.83	3.29
Across industries (F-F Industry)	41,411	25.02	22.42	8	19	37	0.58	0.31	0.32	0.62	0.85	1.45	0.65	1	1.2	1.62
Within industry (GICS Industry)	41,411	16.65	14.19	5	14	25	0.42	0.29	0.17	0.39	0.64	2.65	1.99	1.18	1.9	3.48
Across industries (GICS Industry)	41,411	25.44	22.79	8	19	37	0.58	0.29	0.36	0.61	0.83	1.38	0.56	1	1.17	1.5
Connected firms	23	0.94	0.02	0.92	0.94	0.96										
Indirect peers																
		Number of connections					Connections (%)									
Overall	41,411	414.72	238.07	222	382	576	0.07	0.13	0	0.01	0.09					
Within industry (3 Digit-SIC code)	41,411	20.54	32.72	1	5	25	0.93	0.13	0.91	0.99	1					
Across industries (3 Digit-SIC code)	41,411	394.18	239.32	202	359	555	0.14	0.15	0.03	0.08	0.21					
Within industry (F-F Industry)	41,411	43.36	41.69	12	31	61	0.86	0.15	0.79	0.92	0.97					
Across industries (F-F Industry)	41,411	371.36	235.09	182	335	528	0.09	0.11	0.02	0.05	0.12					
Within industry (GICS Industry)	41,411	25.13	20.32	8	21	38	0.91	0.11	0.88	0.95	0.98					
Across industries (GICS Industry)	41,411	389.59	238.32	196	356	551										

Note. This table presents the descriptive statistics for the analyst coverage network in terms of number of direct and indirect peers, the average number of common analysts by firm-year broken down into within and across industries.

when there is a more than 1% increase in the book value of total debt relative to the book value of total assets, we find that firms issue debt during 36% of the firm-years in our sample period. We use two variables to identify equity issuances. Our first variable defines equity issuances as instances when the difference between cash flow from equity issues less cash flow from equity repurchases is greater than 1% of the book value of total assets, *Net equity*. Based on this definition, firms issue equity in 22% of firm-year. When we define gross equity issuances as years when the cash flow from equity issues is more than 1% of the book value of total assets, *Gross equity*, we find that equity issuances occur in 36% of the firm-years.

Table 2 also provides the summary information for all the control variables in our sample. The summary values are similar to those for the full CRSP-Compustat-IBES merged sample. We winsorize all our variables of interest at the 1st and 99th percentiles.

In Table 3, we provide the summary information for *Equity shock*. Although the average value of the own firm's equity shock ($Equity\ shock^{OWN}$) in our sample is close to zero at -0.03 , it has sufficient dispersion with a standard deviation of 0.49. As expected, the dispersion declines when the equity shock is averaged over either the industry or analyst peer firms.

4. Empirical Results

In this section, we discuss our empirical results. The discussion is divided into four parts. First, we document a positive association between a firm's financial policy and that of its analyst peers. Second, we use *Equity shock* as an exogenous peer firm characteristic to establish the existence of social effects distinct from correlated effects. We then use the friends-of-friends approach to isolate endogenous peer effects from exogenous peer effects. Finally, we provide a series of robustness and placebo tests to distinguish peer effects operating through analyst networks from those operating within industries and perform several cross-sectional tests that provide a richer picture of the mechanism underlying the peer effects.

4.1. Baseline Regressions

We provide the results of estimating Equation (1) in our full sample in Table 4. The outcome variable in columns (1) and (3) is *Leverage* in first difference and level, respectively. The positive and significant coefficient on *Industry average* highlights the positive association between a firm's leverage and average leverage of other firms in its industry (Welch 2004, Frank and Goyal 2008). Coefficients on the firm-specific control variables are consistent with prior studies (Rajan and Zingales 1995). From the coefficients on the industry average characteristics, we find that only industry average *Profitability* is consistently related to firm leverage. Consistent with the findings in

Table 2. Summary Statistics: Outcome and Control Variables

Panel A: Firm-specific variables										
	N	Focal Firms			Industry average (Three-Digit SIC Code)			Peer firm simple average		
		Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Δ Leverage	41,411	0.01	0.1	0	0.01	0.05	0.01	0.01	0.04	0.01
Leverage	41,411	0.22	0.22	0.15	0.23	0.14	0.2	0.21	0.11	0.19
Net debt	41,411	0.36	0.48	0	0.33	0.17	0.31	0.38	0.17	0.37
Net equity	41,411	0.22	0.42	0	0.22	0.15	0.21	0.22	0.18	0.18
Gross equity	41,411	0.36	0.48	0	0.3	0.18	0.3	0.38	0.23	0.33
Log(Sales)	41,411	6.74	1.79	6.64	5.84	1.15	5.7	7.11	0.95	7.16
Market to book	41,411	1.68	1.27	1.28	1.62	0.67	1.44	1.83	0.76	1.66
Profitability	41,411	0.12	0.11	0.13	0.09	0.07	0.1	0.14	0.05	0.14
Tangibility	41,411	0.29	0.23	0.22	0.28	0.19	0.22	0.29	0.16	0.26
Δ Log(Sales)	41,411	0.1	0.22	0.08	0.09	0.11	0.1	0.11	0.1	0.11
Δ Market to book	41,411	−0.04	0.8	0	−0.06	0.39	−0.04	−0.06	0.44	−0.01
Δ Profitability	41,411	0	0.07	0	−0.01	0.03	0	0	0.02	0
Δ Tangibility	41,411	0	0.04	0	0	0.02	0	0	0.01	0
Log(1+Coverage)	41,411	2.16	0.73	2.2	1.92	0.32	1.92	2.42	0.31	2.45

Panel B: Peer firm weighted averages										
	N	Full sample			Same industry			Different industry		
		Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Δ Leverage	41,411	0.01	0.05	0.01	0.01	0.06	0	0.01	0.05	0
Leverage	41,411	0.21	0.12	0.19	0.18	0.17	0.13	0.2	0.11	0.19
Net debt	41,411	0.38	0.18	0.37	0.32	0.3	0.28	0.36	0.2	0.36
Net equity	41,411	0.22	0.18	0.17	0.19	0.25	0.06	0.2	0.19	0.14
Gross equity	41,411	0.38	0.24	0.34	0.34	0.34	0.25	0.35	0.25	0.31
Log(Sales)	41,411	7.28	1.05	7.3	6.23	2.82	6.96	7.09	1.7	7.36
Market to book	41,411	1.86	0.83	1.66	1.61	1.17	1.41	1.72	0.78	1.61
Profitability	41,411	0.14	0.05	0.15	0.12	0.08	0.13	0.14	0.05	0.14
Tangibility	41,411	0.3	0.18	0.26	0.26	0.23	0.18	0.27	0.15	0.25
Δ Log(Sales)	41,411	0.11	0.11	0.11	0.1	0.14	0.09	0.1	0.11	0.1
Δ Market to book	41,411	−0.06	0.46	−0.01	−0.05	0.54	0	−0.05	0.45	0
Δ Profitability	41,411	0	0.03	0	0	0.03	0	0	0.03	0
Δ Tangibility	41,411	0	0.01	0	0	0.02	0	0	0.01	0
Log(1+Coverage)	41,411	2.52	0.36	2.55	2.17	0.99	2.49	2.39	0.58	2.49

Notes. This table reports the summary statistics for the outcome and control variables that we employ in our analysis. All variables used in the regression analysis are winsorized at the 1st and 99th percentiles and are defined in Appendix A.

Leary and Roberts (2014), firms from more profitable industries have higher leverage.

In columns (2) and (4), we augment the model with *Peer average*, the weighted average leverage (in first difference and level) of all firms in the analyst network. We also include the weighted average characteristics of the analyst peer firms in the regressions. We find that the

coefficient on *Peer average* is positive and significant. The coefficient on *Peer average* is significantly larger than that on *Industry average*, and inclusion of *Peer average* reduces the size of the coefficient on *Industry average* in first difference (level) from 0.507 (0.435) to 0.285 (0.302). This is consistent with analyst peer firm leverage having a large effect on a firm's leverage. Focusing on the peer firm

Table 3. Summary Statistics: Equity Shocks

	N	Mean	SD	P25	Median	P75
Equity shock ^{OWN}	41,411	−0.03	0.49	−0.32	−0.09	0.14
Equity shock ^{IND} (three-digit SIC code)	41,411	−0.03	0.16	−0.12	−0.04	0.04
Equity shock ^{ACN} (weighted average)	41,411	−0.04	0.12	−0.1	−0.04	0.02
Equity shock ^{ACN} (simple average)	41,411	−0.03	0.11	−0.09	−0.04	0.02
Equity shock ^{ACN} (indirect peer)	41,411	−0.03	0.05	−0.06	−0.03	−0.01
Equity shock ^{ACN} (indirect peer excl. same industry)	41,411	−0.03	0.05	−0.06	−0.03	−0.01

Notes. This table shows the summary statistics of the equity shock for the individual firms, industry peers; and direct and indirect peers. All variables used in the regression analysis are winsorized at the 1st and 99th percentiles.

Table 4. Baseline Specification I. Analyst Peer Firms vs. Industry Peers

	<i>ΔLeverage</i>		<i>Leverage</i>		<i>Net Equity</i>		<i>Gross Equity</i>		<i>Net Debt</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Peer average		0.547 (0.020)***		0.391 (0.024)***		0.301 (0.022)***		0.278 (0.020)***		0.216 (0.021)***
Industry average	0.507 (0.017)***	0.285 (0.016)***	0.435 (0.022)***	0.302 (0.021)***	0.287 (0.024)***	0.199 (0.023)***	0.321 (0.023)***	0.238 (0.023)***	0.204 (0.022)***	0.152 (0.022)***
Own characteristics										
Log(Sales)	0.026 (0.003)***	0.021 (0.003)***	0.040 (0.004)***	0.039 (0.004)***	−0.092 (0.008)***	−0.090 (0.008)***	−0.075 (0.009)***	−0.075 (0.009)***	−0.053 (0.008)***	−0.055 (0.009)***
Market to book	0.001 (0.0005)***	0.002 (0.0005)***	−0.019 (0.001)***	−0.018 (0.001)***	0.064 (0.004)***	0.061 (0.004)***	0.068 (0.004)***	0.067 (0.004)***	0.010 (0.004)**	0.009 (0.004)**
Profitability	−0.028 (0.009)***	−0.023 (0.009)**	−0.299 (0.018)***	−0.294 (0.018)***	−0.167 (0.041)***	−0.170 (0.042)***	0.159 (0.042)***	0.159 (0.042)***	0.334 (0.046)***	0.318 (0.046)***
Tangibility	0.093 (0.013)***	0.087 (0.013)***	0.069 (0.024)***	0.071 (0.023)***	0.100 (0.044)**	0.103 (0.043)**	0.074 (0.047)	0.073 (0.047)	0.299 (0.047)***	0.305 (0.047)***
Log(1+Coverage)	0.008 (0.0009)***	0.008 (0.001)***	0.001 (0.003)	−0.003 (0.003)	−0.056 (0.007)***	−0.051 (0.007)***	−0.048 (0.008)***	−0.041 (0.008)***	0.021 (0.009)**	0.022 (0.009)**
Log(1+Connections)	−0.005 (0.0009)***	−0.003 (0.0009)***	−0.016 (0.003)***	−0.016 (0.003)***	0.036 (0.006)***	0.040 (0.006)***	0.040 (0.006)***	0.040 (0.006)***	0.028 (0.007)***	0.029 (0.007)***
Peer characteristics										
Log(Sales)		0.013 (0.007)*		−0.011 (0.003)***		0.012 (0.007)*		0.020 (0.007)***		−0.0009 (0.008)
Market to book		−0.004 (0.001)***		0.005 (0.002)**		0.0009 (0.007)		−0.013 (0.007)*		0.0006 (0.007)
Profitability		−0.036 (0.026)		0.078 (0.036)**		0.063 (0.089)		−0.075 (0.091)		0.086 (0.101)
Tangibility		0.036 (0.043)		−0.060 (0.024)**		0.024 (0.049)		0.056 (0.053)		−0.080 (0.060)
Log(1+Coverage)		−0.004 (0.002)**		0.018 (0.006)***		−0.031 (0.014)**		−0.034 (0.016)**		−0.007 (0.018)
Industry characteristics										
Log(Sales)	−0.003 (0.006)	−0.007 (0.006)	−0.010 (0.005)**	−0.008 (0.005)*	0.009 (0.009)	0.005 (0.009)	−0.003 (0.011)	−0.006 (0.011)	−0.008 (0.012)	−0.007 (0.012)
Market to book	−0.006 (0.002)***	−0.003 (0.002)	−0.005 (0.003)	−0.005 (0.003)	0.014 (0.007)**	0.006 (0.008)	0.011 (0.008)	0.012 (0.009)	0.017 (0.008)**	0.016 (0.009)*
Profitability	0.031 (0.020)	0.024 (0.022)	0.135 (0.028)***	0.114 (0.029)***	−0.036 (0.067)	−0.017 (0.070)	−0.117 (0.076)	−0.098 (0.079)	0.120 (0.079)	0.052 (0.082)
Tangibility	−0.076 (0.036)**	−0.068 (0.037)*	0.049 (0.043)	0.036 (0.042)	0.105 (0.074)	0.070 (0.073)	0.116 (0.088)	0.080 (0.087)	−0.055 (0.098)	−0.019 (0.098)
Log(1+Coverage)	0.006 (0.002)***	0.005 (0.002)**	0.006 (0.007)	0.003 (0.007)	−0.047 (0.013)***	−0.031 (0.013)**	−0.061 (0.015)***	−0.047 (0.015)***	0.001 (0.017)	0.004 (0.017)
Observations	41,411	41,411	41,411	41,411	41,411	41,411	41,411	41,411	41,411	41,411
R ²	0.153	0.179	0.775	0.781	0.394	0.399	0.457	0.461	0.234	0.237

Notes. The table presents the OLS coefficients for the following regression:

$$y_{ijt} = \alpha + \beta_1 y_{-it}^{ACN} + \beta_2 y_{-ijt}^{IND} + \gamma_1' X_{-it-1}^{ACN} + \gamma_2' X_{-ijt-1}^{IND} + \gamma_3' X_{ijt-1} + \delta' u_i + \phi' v_t + \epsilon_{ijt}.$$

The outcome variables are *ΔLeverage* (columns (1) and (2)), *Leverage* (columns (3) and (4)), *Net Equity* (columns (5) and (6)), *Gross Equity* (columns (7) and (8)), and *Net Debt* (columns (9) and (10)). The main independent variables are *Peer average* and *Industry average*, which measure the contemporaneous averages of the respective outcome variable for analyst and industry peers, respectively. All other control variables are lagged one period, except the number of firm’s connections. In the specification with *ΔLeverage* as the dependent variable, all the control variables are in first difference form, except the number of firm’s connections and analyst coverage; and we include year and industry fixed effects (three-digit SIC code). The remaining specifications include firm and year fixed effects. Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity, we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

characteristics, we find that only the coefficients on peer firm average *Log(Sales)*, *Market to book* and *analyst coverage* are significant in both columns.

In columns (5) to (8), we focus on equity issuances and, irrespective of our measure of equity issuance, we find that there is a positive association between equity

issuances by a firm and equity issuances by analyst peer firms in the same year. The coefficients on both *Peer average* and *Industry average* are of similar magnitude. In columns (9) and (10), we repeat our tests with *Net debt* as the dependent variable, and from column (10), we find that there is a positive association between the probability of

debt issuances by a firm in a year and debt issuances of analyst-connected peer firms. Here again we find that the coefficient on *Peer average* is larger than that on *Industry average*. Interestingly, we find that none of the industry or analyst peer characteristics are significantly related to a firm's decision to issue debt. Overall, our results in Table 4 show that firm financial policies are positively related to the financial policies of firms that are connected through common analysts. The magnitude of the association is greater than that between firm financial policy and industry average financial policies.

In Table IA-2 of the online appendix, we differentiate between analyst peers that are from the same industry and those that are from different industries to see if these two groups have a similar effect on firm financial decisions. We do this by replacing *Peer average* with two variables: *Peer average (same industry)* and *Peer average (different industry)*. These are the weighted averages of the outcome variable for same and different industry analyst peers. We calculate the weighted average using the methodology outlined in Section 3. From columns (1) and (2) of Table IA-2, we find that the coefficients on both same and different industry peer averages are positive and significant. The coefficients are also of similar size. This indicates that firm leverage is related similarly to the leverage of analyst peers from both the same and different industries. In unreported tests we find that the coefficients on same and different industry peer averages are not statistically distinguishable. These results further reinforce the conclusion that the analyst network may have an independent effect on firm leverage apart from the industry effect documented in Leary and Roberts (2014). We find similar results for net debt issuance and both net and gross equity issuance. It is noteworthy that the different industry analyst peers have a larger influence on a firm's decision to issue equity as compared with same industry analyst peers.

4.2. Reduced Form and Structural Regressions

Having established a positive association between analyst peers' financial policies and a firm's own financial policy, we now go to our next set of tests wherein we use *Equity shock* as an exogenous peer firm characteristic to control for correlated effects. In Table 5, we report the results of a reduced form estimation wherein we include *Equity shock^{ACN}* and *Equity shock^{IND}* instead of peer and industry average financial policy and repeat our tests.¹⁷ We perform the reduced form analysis to provide evidence of social effects (endogenous or exogenous). However, as discussed previously, this specification cannot distinguish endogenous from exogenous peer effects. In this table, we also include *Equity shock^{IND}* to highlight that the effect of *Equity shock^{ACN}* is robust to controlling for industry characteristics, suggesting that our peer effect results are not only due to peer firms from the same industry. We explore this issue further in subsequent tests.

From columns (1) and (2), we find that all three equity shock variables (lagged one period), *Equity shock^{OWN}*, *Equity shock^{IND}*, and *Equity shock^{ACN}* are negatively associated with a firm's market leverage (first difference and level). The negative and significant coefficient on *Equity shock^{ACN}* is consistent with the presence of social effects within the analyst network. When we model leverage (column 2), our coefficient estimates on *Equity shock^{IND}* and *Equity shock^{OWN}* are similar to those reported in Leary and Roberts (2014) (table 4). In the change specification (column (1)), however, the industry average shock becomes statistically insignificant once we control for *Equity shock^{ACN}*.

Columns (3) and (4) indicate a strong positive association between *Equity shock^{ACN}* in a year and the probability of a firm making equity issues the next year. This suggests that the leverage results in the first two columns are driven, at least in part, by the influence of analyst network peers on firms' equity issuance decisions. In column (5), we estimate a similar specification for net debt issuances. Consistent with the leverage and equity issuance results, we find a negative coefficient on the peer firm equity shock, although the statistical significance is weak. In addition, we find an insignificant relation between *Net Debt* and the firm's own equity shock, which makes the coefficient on the peer average equity shock difficult to interpret. The results in column (6) indicate that this weak relation between debt issues and equity shocks is due largely to the comparison set. That is, column (5) compares observations with a debt issuance to all other observations. In column (6), we restrict the sample to periods of active financing decisions, that is, where there is either a debt issuance or an equity issuance, to test for the influence of peer firms on the debt-equity choice. The dependent variable is an indicator equal to one (zero) if the issuance is debt (equity). The results indicate that, conditional on some form of issuance, the choice between debt and equity is strongly related both to the firm's own equity shock and that of its analyst network peers.¹⁸ Overall, our evidence in Table 5 shows that there appear to be strong social effects within analyst networks for leverage, equity issuance, and security choice decisions.

Agency-based considerations may play a role in managerial responses to peer effects, because managers may prefer leverage-decreasing versus leverage-increasing decisions to reduce the disciplining role of debt (Jensen 1986, Hart and Moore 1994, Novaes 2002). In Table IA-3 in the online appendix, we test whether we observe any difference in the economic significance of the peer effect between positive and negative peer equity shocks. We add a dummy variable, *Positive*, when peers' idiosyncratic returns are positive. Our results provide no evidence of relative differences between leverage-increasing and leverage-decreasing peer effects, suggesting that agency conflicts may not be a first-order driver of firms' responses to their peers.

Table 5. Reduced Form Regression with Idiosyncratic Equity Shock

	$\Delta\text{Leverage}$ (1)	Leverage (2)	Net Equity (3)	Gross Equity (4)	Net Debt (5)	$\text{Net Debt (Issuer sample)}$ (6)
$\text{Equity shock}^{\text{ACN}}$	−0.026 (0.005)***	−0.024 (0.006)***	0.056 (0.019)***	0.079 (0.019)***	−0.039 (0.023)*	−0.099 (0.030)***
$\text{Equity shock}^{\text{IND}}$	−0.004 (0.004)	−0.020 (0.004)***	0.002 (0.013)	0.019 (0.014)	0.007 (0.017)	−0.024 (0.023)
$\text{Equity shock}^{\text{OWN}}$	−0.006 (0.001)***	−0.017 (0.002)***	0.057 (0.005)***	0.072 (0.005)***	−0.006 (0.005)	−0.062 (0.007)***
Own characteristics						
Log(Sales)	0.025 (0.003)***	0.042 (0.004)***	−0.090 (0.008)***	−0.075 (0.009)***	−0.053 (0.009)***	0.034 (0.012)***
Market to book	0.003 (0.0006)***	−0.016 (0.001)***	0.056 (0.004)***	0.060 (0.004)***	0.010 (0.004)**	−0.028 (0.005)***
Profitability	−0.023 (0.009)**	−0.295 (0.018)***	−0.179 (0.042)***	0.142 (0.043)***	0.327 (0.046)***	0.061 (0.060)
Tangibility	0.085 (0.013)***	0.060 (0.024)**	0.121 (0.044)***	0.092 (0.047)*	0.300 (0.047)***	0.085 (0.068)
Log(1+Coverage)	0.009 (0.001)***	−0.004 (0.003)	−0.038 (0.008)***	−0.027 (0.008)***	0.022 (0.009)**	0.041 (0.012)***
$\text{Log(1+Connections)}$	−0.004 (0.0009)***	−0.016 (0.003)***	0.036 (0.006)***	0.038 (0.006)***	0.029 (0.007)***	−0.016 (0.009)*
Peer characteristics						
Log(Sales)	0.043 (0.007)***	0.0005 (0.003)	−0.005 (0.007)	0.003 (0.007)	0.003 (0.008)	−0.006 (0.011)
Market to book	−0.004 (0.002)***	−0.010 (0.002)***	0.021 (0.007)***	0.007 (0.007)	0.002 (0.007)	−0.009 (0.009)
Profitability	−0.059 (0.028)**	−0.099 (0.037)***	−0.083 (0.088)	−0.068 (0.091)	0.154 (0.101)	0.100 (0.137)
Tangibility	0.125 (0.044)***	0.042 (0.024)*	0.037 (0.049)	0.027 (0.053)	−0.041 (0.060)	−0.010 (0.080)
Log(1+Coverage)	−0.004 (0.002)**	0.003 (0.007)	−0.029 (0.015)*	−0.015 (0.015)	−0.014 (0.018)	−0.005 (0.023)
Industry characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Observations	41,411	41,411	41,411	41,411	41,411	20,150
R^2	0.118	0.768	0.395	0.458	0.231	0.613

Notes. The table presents the results of the following OLS regression:

$$y_{ijt} = \alpha_0 + \alpha_1 \text{Equity shock}^{\text{ACN}}_{-it-1} + \alpha_2 \text{Equity shock}^{\text{IND}}_{-ijt-1} + \alpha_3 \text{Equity shock}^{\text{OWN}}_{ijt-1} + \gamma'_1 X^{\text{ACN}}_{-it-1} + \gamma'_2 X^{\text{IND}}_{-ijt-1} + \gamma'_3 X_{ijt-1} + \delta' u_i + \phi' v_t + \epsilon_{ijt}.$$

The outcome variables are $\Delta\text{Leverage}$ (column (1)), Leverage (column (2)), Net Equity (column (3)), Gross Equity (column (4)), and Net Debt (column (5)). Column (6) isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). The main independent variables are $\text{Equity shock}^{\text{ACN}}$ and $\text{Equity shock}^{\text{IND}}$ which measure the lagged averages of the idiosyncratic equity shock for analyst peers and industry peers, respectively. All control variables are lagged one period, except the number of firm's connections. In the specification with $\Delta\text{Leverage}$ as the dependent variable, all the control variables are in first difference form, except the number of firm's connections and analyst coverage; and we include year and industry fixed effects (three-digit SIC code). The remaining specifications include firm and year fixed effects. Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

In Table IA-4 of the online appendix, we use alternate thresholds to define the net (gross) equity issuance dummy (1%, 3%, and 5% of total assets). In all cases, we find that our results are robust to using different thresholds to identify net and gross equity issuance.

Once we include $\text{Equity shock}^{\text{ACN}}$, the coefficients on $\text{Equity shock}^{\text{IND}}$ are insignificant in all the columns. $\text{Equity shock}^{\text{ACN}}$ is different from $\text{Equity shock}^{\text{IND}}$ along two dimensions. First, it averages across firms connected through common analysts irrespective of their industry affiliation. Second, it is a weighted average with the weights equal to the number of common analysts. To see which of these is responsible for $\text{Equity shock}^{\text{IND}}$ losing

statistical significance, in unreported tests, we repeat the estimation after replacing $\text{Equity shock}^{\text{ACN}}$ with $\text{Equity shock}^{\text{ACN}}_{\text{simple average}}$. We find that the coefficients on $\text{Equity shock}^{\text{IND}}$ continue to be insignificant (and the coefficient on $\text{Equity shock}^{\text{ACN}}_{\text{simple average}}$ significantly positive) in that specification. This highlights that it is the fact that $\text{Equity shock}^{\text{ACN}}$ averages over a specific set of peers that is responsible for soaking up the effect of $\text{Equity shock}^{\text{IND}}$.

SEO decisions are another corporate financial policy that can be influenced by peer effects. The influence of analysts on equity issuance decisions has been previously documented: Chang et al. (2006) find that firms

covered by fewer analysts are less likely to issue equity, and Bowen et al. (2008) and Chan and Chan (2014) find that analyst coverage can mitigate the information asymmetry associated with SEO discounts. In Table 6, we investigate peer effects on SEOs. Specifically, we test whether a firm is more likely to have an SEO event when peer firms have positive idiosyncratic returns in the previous year and whether SEOs that follow peers' decisions are associated with better announcement returns.

We obtain SEO events from Securities Data Company (SDC) Platinum and use a linear probability model with the same econometric specification for equity issuances used in Table 5. The dependent variable in our panel data analysis is an indicator variable, SEO, that takes the value of one when a firm has an SEO in a given year and zero otherwise. In column (1), we find that peer equity shocks are positively associated with future SEOs of firms; higher idiosyncratic returns of analyst coverage network peer firms increase the probability that the focal firm issues new seasoned equity in the next year.

In columns (2) to (4), we analyze SEO announcement returns using raw returns, market-adjusted returns, and buy and hold abnormal returns, respectively, within a $\pm$ three-day window around the filing date. Across all three specifications, we find that equity shocks of peer firms are positively and significantly associated with SEO announcement returns. These results suggest, first, that firms are influenced by their analyst network peers also when making SEO decisions. Furthermore, SEOs that follow those of analyst network peers appear to be associated with higher announcement returns, consistent with previous evidence that analysts reduce

information asymmetry around security issuances and of information spillovers across firms in the analyst's coverage network (Chan and Chan 2014).

In Table 7, we provide the results of the two-stage least squares estimation that uses *Equity shock*^{ACN} as an instrument for the average financial policies of peer firms. In all the specifications we also include the average financial policies of firms in the same industry as an additional control. At the top of Table 7, we provide the coefficients on the instruments from the first stage regression. Estimating the 2SLS has advantages and disadvantages relative to the reduced form. The advantage is that it allows us to estimate the magnitude of the impact of analyst peer firm policies on firms' financial decisions. The limitation, however, is that interpreting the magnitude in this way requires us to assume that the peer firms' equity shock influences firm *i*'s financial policy only through its effect on peers' financial policies. As discussed earlier, it is possible that peers' equity shock influences firm *i*'s policies because it is a shock to the peers' characteristics, such as investment opportunities or competitive position. This would represent an exogenous peer effect, in which case we would be wrong to attribute the entire magnitude to endogenous peer effects, that is, the effect of peers' policies on firm *i*'s policies.

Despite this caveat, the results in Table 7 are instructive. The first stage results indicate that *Equity shock*^{ACN} is significantly related to peer firm leverage (columns (1) and (2)) and equity issuance (columns (3) and (4)) decisions. Furthermore, the *F* values for weak instrument tests shown at the bottom of the table are all large and greater than the threshold of 10.

Table 6. SEOs and Idiosyncratic Equity Shock

	SEO (1)	CRET[-3,3] (2)	CAR[-3,3] (3)	BHAR[-3,3] (4)
<i>Equity shock</i> ^{ACN}	0.028 (0.010)***	0.050 (0.023)**	0.045 (0.022)**	0.045 (0.021)**
<i>Equity shock</i> ^{IND}	0.007 (0.008)	0.005 (0.019)	-0.0003 (0.018)	-0.0008 (0.018)
<i>Equity shock</i> ^{OWN}	0.028 (0.003)***	0.003 (0.004)	-0.006 (0.003)*	-0.006 (0.003)*
Own characteristics	Yes	Yes	Yes	Yes
Peer characteristics	Yes	Yes	Yes	Yes
Industry characteristics	Yes	Yes	Yes	Yes
Observations	41,411	2,341	2,341	2,341
<i>R</i> ²	0.186	0.097	0.105	0.106
Mean SEO announcement returns		-0.008***	-0.021***	-0.022***

Notes. The table presents the results of the effect of peer equity shocks on the probability of a SEO and the influence on the SEO announcement returns. The outcome variables are SEO (column (1)), which is a dummy that takes the value of one if a firm in a given year had a SEO, and it is zero otherwise. CRET[-3,3] (CAR[-3,3]) and BHAR[-3,3] are the cumulative raw (abnormal) returns and buy-and-hold abnormal returns, respectively, in a three-day window surrounding the SEO announcement (filing) day. The main independent variables are *Equity shock*^{ACN} and *Equity shock*^{IND}, which measure the lagged averages of the idiosyncratic equity shock for analyst peers and industry peers, respectively. All control variables are lagged one period, except the number of firm's connections. In column (1) standard errors are clustered at the firm level, whereas columns (2) to (4) are standard errors double clustered at the industry level (three-digit SIC code) and year level. Column (1) includes firm and year fixed effects. Columns (2) to (4) include industry and year fixed effects. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

Table 7. Two-Stage Least-Square Instrumentals Variable Regression Using Equity Shock

	Δ Leverage (1)	Leverage (2)	Net Equity (3)	Gross Equity (4)	Net Debt (Issuer sample) (5)
First stage					
Equity shock ^{ACN}	−0.013 (0.002)***	−0.020 (0.003)***	0.078 (0.007)***	0.113 (0.008)***	−0.040 (0.011)***
Second stage					
Instrumented peer average	1.723 (0.410)***	1.043 (0.302)***	0.672 (0.224)***	0.660 (0.163)***	2.117 (0.842)**
Industry average	−0.185 (0.165)	0.086 (0.100)	0.095 (0.066)	0.122 (0.052)**	−0.020 (0.111)
Own characteristics					
Equity shock ^{OWN}	−0.005 (0.001)***	−0.017 (0.001)***	0.057 (0.004)***	0.072 (0.004)***	−0.060 (0.007)***
Log(Sales)	0.017 (0.003)***	0.037 (0.004)***	−0.088 (0.008)***	−0.074 (0.008)***	0.014 (0.013)
Market to book	0.004 (0.0007)***	−0.016 (0.001)***	0.054 (0.004)***	0.059 (0.004)***	−0.028 (0.004)***
Profitability	−0.010 (0.010)	−0.289 (0.017)***	−0.196 (0.039)***	0.133 (0.040)***	0.045 (0.058)
Tangibility	0.074 (0.014)***	0.078 (0.022)***	0.115 (0.040)***	0.087 (0.044)*	0.124 (0.072)*
Log(1+Coverage)	0.007 (0.001)***	−0.012 (0.004)***	−0.033 (0.007)***	−0.014 (0.008)*	0.023 (0.014)
Log(1+Connections)	−0.002 (0.001)**	−0.013 (0.003)***	0.040 (0.006)***	0.035 (0.006)***	−0.014 (0.011)
Peer characteristics	Yes	Yes	Yes	Yes	Yes
Industry characteristics	Yes	Yes	Yes	Yes	Yes
Observations	41,411	41,411	41,411	41,411	19,109
Kleibergen-Paap F value	52.651	53.391	112.661	208.026	12.801
Cragg-Donald F value	88.724	90.738	250.692	410.395	23.639
Anderson-Rubin F value	21.398	14.1	9.313	17.348	10.853
Anderson-Rubin p value	3.83 e-06	0.0002	0.002	0.00003	0.001

Notes. The table presents the results of the 2SLS regression that relates peer firm average financial policy to firm financial policy. The outcome variables are Δ Leverage (column (1)), Leverage (column (2)), Net Equity (column (3)), and Gross Equity (column (4)). Column (5) isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). The main endogenous variables are peer averages of the outcome variables. The instrument in the first stage is *Equity shock^{ACN}*, which measures the lagged averages of the idiosyncratic equity shock for analyst peers. All control variables are lagged one period, except the number of firm’s connections. In the specification with Δ Leverage as the dependent variable all the control variable sare in first difference form, except the number of firm’s connections and analyst coverage; and we include year and industry fixed effects (three-digit SIC code). The remaining specifications include firm and year fixed effects. Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity, we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

Focusing on the results of the second stage, we find that the coefficient on the instrumented peer average leverage is positive and significant in columns (1) and (2), consistent with the presence of peer effects in leverage decisions that propagate through analyst networks. Our estimates are also economically significant. The coefficient on *Peer average* in column (2) indicates that a 1-standard-deviation increase in peer firm weighted average leverage is associated with a 0.569-standard-deviation increase in the firm’s leverage ($0.569 = 1.043 \times (0.12/0.22)$). However, this is only a partial effect. In a linear model of social interactions, any impact on firm *i*’s leverage will feed back to that of its peers (given a non-zero β_1 in Equation (1)). In the presence of these feedback effects, the coefficients on y_{-it} (and X_{-ijt}) do not necessarily give the full marginal effect of a change in the independent variable on the outcome variable (Kline and

Tamer 2014, Leary and Roberts 2014). Nonetheless, we can get a sense of the relative magnitude of different factors by comparing scaled coefficients across the independent variables. By this measure, the leverage decisions of network peers seems to have a larger impact on firms’ own leverage choice than the standard included firm characteristics. *Log(Sales)* is closest in magnitude, with a scaled coefficient of 0.3.

From columns (3) and (4), we find that the decision of peer firms to issue equity in a year is associated with the own firm’s decision to issue equity. We find that the effect of analyst peers is greater than the effect of industry peers. Our estimates are also economically significant. Scaling by the relevant standard deviations, the coefficients on instrumented peer equity issuance are large relative to those on the included firm characteristics (only firm size has a larger scaled coefficient in the net

equity specification). Results in column (5) similarly find a significant positive relation between the debt-equity choice of analyst network peers (conditional on an issuance) and a firm's own debt-equity choice.

4.3. Indirect Peer Approach

In Table 8, we use the friends-of-friends methodology to help isolate exogenous peer effects from endogenous peer effects. Specifically, we identify indirect peer firms for every firm. These are firms that are not directly connected to a firm through common analysts but are connected to one or more of its analyst network peers. We then estimate a two-stage least squares model in which we use the average equity shock of these “indirect peers” as an instrument for the financial policies of a firm's direct peers to identify endogenous peer effects in financial policy.¹⁹

As discussed in Section 3, to separate contextual from endogenous peer effects, the key identification assumption is that the characteristics of the indirect peers used as

instruments are uncorrelated with the characteristics of the direct peers. This is likely to be true for idiosyncratic return shocks as they isolate value-relevant events that are unique to the indirect peers (i.e., this is the identification assumption used in the previous section).

The first row of Table 8 presents the coefficients on the indirect peer average equity shock from the first stage. We find that the equity shocks of indirect peer firms are significantly related to the level and change in leverage, net and gross equity issuances, and the debt-equity choice of direct peers. Furthermore, the *F* values indicate that for these policy variables the instrument easily passes the weak instrument test.

Regarding the second stage, in the odd numbered columns, which include all indirect peers, we find a significant relation between a firm's financial policies and those of their direct peers for both the change and the level of leverage, gross equity issuances, and the debt-equity choice (though the coefficient on the instrumented average peer debt-equity choice is significant only at

Table 8. Two-Stage Least Square Instrumentals Variable Regression Using Indirect Peers

Indirect peers	All Δ Leverage	Diff. Ind. Leverage	All Leverage	Diff. Ind. Leverage	All Net equity	Diff. Ind. Net equity	All Gross equity	Diff. Ind. Gross equity	All Net debt (Issuer sample)	Diff. Ind. Net debt (Issuer sample)
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
First stage										
Equity shock ^{ACN} (indirect peers)	−0.038 (0.007)***	−0.041 (0.007)***	−0.077 (0.011)***	−0.060 (0.011)***	0.096 (0.026)***	0.107 (0.027)***	0.170 (0.028)***	0.144 (0.028)***	−0.119 (0.041)***	−0.136 (0.039)***
Second stage										
Instrumented peer average	0.914 (0.418)**	1.110 (0.375)***	0.964 (0.274)***	1.019 (0.339)***	0.517 (0.668)	0.499 (0.587)	1.083 (0.412)***	1.054 (0.474)**	1.530 (0.917)*	1.384 (0.749)*
Industry average (no overlap)	0.069 (0.059)	0.042 (0.053)	0.037 (0.014)***	0.034 (0.017)**	0.045 (0.024)*	0.045 (0.022)**	0.036 (0.016)**	0.036 (0.017)**	0.039 (0.026)	0.042 (0.023)*
Peer average stock return	−0.002 (0.004)	−0.003 (0.004)	0.002 (0.008)	0.004 (0.010)	0.019 (0.060)	0.020 (0.053)	−0.026 (0.042)	−0.024 (0.048)	−0.055 (0.021)***	−0.053 (0.019)***
Own characteristics (incl. equity shock)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Peer characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind. characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	41,411	41,411	41,411	41,411	41,411	41,411	41,411	41,411	16,293	16,293
Kleibergen-Paap <i>F</i> -value	34.032	39.157	52.06	31.411	13.779	15.435	35.447	25.577	8.404	12.081
Cragg-Donald <i>F</i> -value	63.974	78.905	96.329	62.595	32.953	43.272	81.166	62.740	17.077	24.586
Anderson-Rubin <i>F</i> -value	4.56	8.788	13.497	10.722	0.614	0.794	7.879	5.927	3.395	4.027
Anderson-Rubin <i>p</i> -value	0.033	0.003	0.0002	0.001	0.433	0.373	0.005	0.015	0.065	0.045

Notes. The table presents the results of the 2SLS regression that relates peer firm average financial policy to firm capital structure. The outcome variables are Δ Leverage (columns (1) and (2)), Leverage (columns (3) and (4)), Net Equity (columns (5) and (6)) and Gross Equity (columns (7) and (8)). Columns (9) and (10) isolate the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). The main endogenous variables are peer averages of the outcome variables. The instrument in the first stage is *Equity shock^{ACN}* (indirect peers) which measures the lagged averages of the idiosyncratic equity shock for indirect analyst peers. The odd columns show the results of the indirect peer approach using the *Equity shock* of all indirect peers (All). Even columns show the results of the indirect peer approach using the *Equity shock* of indirect peers that are only in different industries as firm *i* (Diff. Ind.). All control variables, but excluding *Industry average (no overlap)* and the number of firm's connections, are lagged one period. In the specification with Δ Leverage as the dependent variable all the control variables are in first difference form, except the number of firm's connections and analyst coverage; and we include year and industry fixed effects (three-digit SIC code). The remaining specifications include firm and year fixed effects. Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

the 10% level). The positive and significant coefficients on *Peer average* for those corporate policies suggest that the average financial policy choice of analyst peer firms has a causal effect on a firm's financial policy choice. Our results are also economically significant. From the coefficient in column (2), we find that a 1-standard-deviation increase in peer firm average leverage is associated with an initial 0.482-standard-deviation increase in a firm's leverage ($0.482 = 0.964 \times (0.11/0.22)$). For equity issuances, we find that a 1-standard-deviation increase in gross equity issuance likelihood by peers leads to a 0.520-standard-deviation increase in gross equity issuance likelihood.

In the even numbered columns, we repeat our tests using an instrument that excludes all indirect peers in the same industry as firm *i*. In these columns, we find similar peer effects for the change and level in firm leverage, gross equity issuance, and the debt-equity choice. Although the coefficient for the level of net equity issuances is insignificant, it is of similar magnitude as that in column (5). Overall, these results suggest that our results are robust to excluding same-industry indirect peers.

It is important to remark that the coefficients associated with industry averages of the outcome variables are also positive and statistically significant, but they are substantially smaller in comparison with peer firm endogenous variables. Our results thus suggest that analyst networks are likely an important source for industry peer effects.

One concern with the test in Table 8 is that the number of indirect peers in each group is considerably larger than the number of direct peers (or industry peers). This diminishes variation in the average equity shock across indirect peer groups. Although the results in Table 8 indicate that enough power remains to identify the endogenous peer effects, in Table IA-5 of the online appendix, we report results of a robustness check to address this concern. Specifically, we limit the set of indirect peers to those with at least two (and four) analysts in common with a direct peer (while still imposing that they have no analysts in common with the firm in question). This limits the size of the indirect peer groups and produces cross-group dispersion in average equity shock only slightly below that of the direct peer groups. The results are similar to those in Table 8, except for the case of leverage in levels (we restricted our sample to include only firms with nonmissing indirect peers' equity shock). In the first stage, we find a considerable reduction in the size of the coefficient on the indirect peers' equity shock when the dependent variable is the level of leverage, likely because we are now relying on only a subset of the peers of the direct peers. However, we continue to find strong first-stage coefficients for the other three policy variables. We do not display the results for net equity issuances as we do not find results

of social effects in Table 8. In the second stage, we now find a significant relation between the instrumented peer average policy and own-firm policies for the change in leverage, as well as for gross equity issues. Importantly, these results are consistent whether we consider at least two (columns (4) to (6)) or four (columns (7) to (9)) analysts in common with a direct peer to identify indirect peers.²⁰ Overall, these results support the findings in Table 8 of endogenous peer effects working through analyst networks and show that these results are robust to varying definitions of indirect peers.

4.4. Role of Analysts in Peer Effects

Results in the previous section provide evidence for endogenous peer effects in analyst coverage networks. Peer effects may be present among firms with common analysts for several potential reasons. For one, these firms are often in the same or similar industries or share other economic linkages (e.g., suppliers and customers). As such, they may be the firms that peers are most likely to look to for comparison on matters such as financial policy (Kaustia and Rantala 2015, 2021). Additionally, analysts may play a more direct role in facilitating peer effects through their role as information intermediaries. Analysts are constantly engaging in two-way communication with firms in their coverage network. Recent research (Martens and Sextroh 2021) suggests that this communication helps transmit information across firms and not just between firms and investors. To the extent that some of this information is relevant to financial decision making, analysts may play a role in propagating financial policies across firms. In this section, we show evidence consistent with analyst information transmission as one mechanism behind our documented peer effects. We begin by showing that the peer effects in analyst coverage networks are distinct from those in industries. That is, the peer effects we document do not arise only because firms with common analysts often are in the same industry (Leary and Roberts 2014). Rather, the analyst coverage network itself is a particular setting in which peer effects are prevalent. We then show cross-sectional variation in the strength of peer effects that is consistent with analysts playing a role in transmitting information across firms. On both fronts, our strategy is to compare estimated peer effects across subsets of firms within the analyst network. Because there is likely to be a large degree of overlap between the sets of indirect peers of these partitions of direct peers, we return for these tests to the specification in Table 5, which uses the idiosyncratic return shocks of direct peers as instruments for their financial policy choices.

4.4.1. Distinguishing Analyst Network Effects from Industry Peer Effects.

Although we control for industry averages in all our tests and show that the effects are present for analyst peers not from the same industry (Table

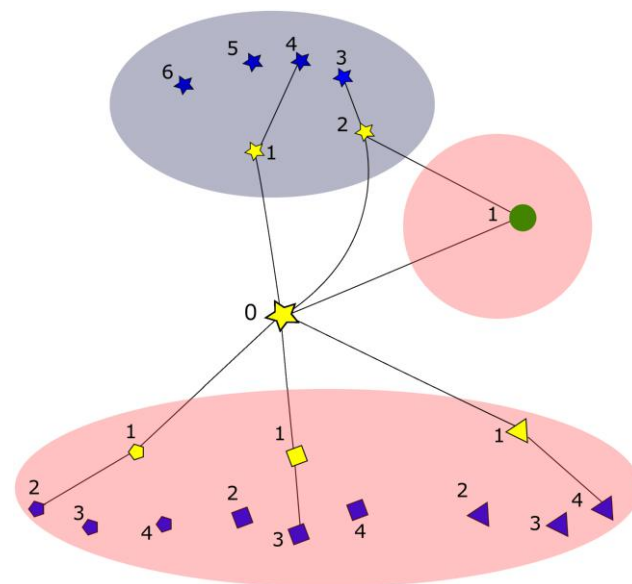
IA-2 in the online appendix), this still raises the question of whether analyst network effects that we document are simply capturing industry peer effects. Our control for industry averages may prove inadequate if the number of analysts in common (which we use to form our weighted average peer equity shock) between pairs of firms in the same industry is higher in comparison with pairs of firms across industries. To the extent that firms in both the same industry and analyst network are more similar and more influential, our analyst peer weighted average may be a more precise measure of industry effects than the simple industry average. We therefore perform several additional tests to address this issue.

We start by re-estimating the regressions from Table 5, but partitioning the set of peer firms to three different groups: (i) firms that are both in the same industry and in the analyst network; (ii) firms that are in the analyst network but not in the same industry; and (iii) firms that are in the same industry but not in the analyst network.²¹ The results, tabulated in the online appendix (Table IA-6), show significant evidence of social interaction effects among all three groups of firms and, somewhat surprisingly, even stronger effects among analyst network only firms than among firms in the same industry. These results suggest that the peer effects operating through analyst networks do not simply reflect industry peer effects.

A potential limitation with the previous test is that analysts may choose to cover firms that are economically connected, even if not in the same industry. For instance, analysts might choose firms in other industries but connected through customer-supplier relationships. Thus, firms that are in the same analyst network, but in different industries, may exert influence on one another as a result of their product market connections rather than the analyst connection. In other words, the connection that an analyst creates between firms may proxy for economic linkages between those firms that as researchers we cannot perfectly observe.

We address this concern in Table 9 by performing a placebo test. Instead of using the average equity shock of firms in the same analyst network, we define a set of pseudo peers that are in the same industry as the peer firms in the analyst network, but do not share a common analyst with firm i . To illustrate, refer to Figure 1, in which numbered shapes represent firms, the shapes themselves (triangle, circle, etc.) represent an industry, and the lines connecting the shapes represent common analysts. Thus, *circle-1*, *pentagon-1*, *square-1*, and *triangle-1* represent firms that are connected to firm *star-0* through common analysts but are in a different industry. To conduct our placebo test, we focus on the firms in the same industry as these firms but that do not have a common analyst with *star-0*. These are firms *pentagon-2* to *pentagon-4*, *square-2* to *square-4*, and *triangle-2* to *triangle-4*. We refer to this average as the $Equity\ shock^{ACN}(pseudo-peer)$ and repeat our tests with this simple average. If the

Figure 1. (Color online) Example of Analyst Coverage Network



Notes. In this figure, we present a hypothetical analyst network for illustration purposes. In the figure, the shape-families represent industries, whereas the shapes represent individual firms. The lines connecting the shapes represent common analysts.

analyst network captures links across firms in different industries, then we should expect $Equity\ shock^{ACN}(pseudo-peer)$ to be significantly related to the corporate policies of the focal firm.

The results in Panel A of Table 9 show that there is no significant relationship between the equity shocks of pseudo-peers and a firm's financial policy. We repeat the tests with alternate industry definitions, including two-digit SIC codes (Panel B), GICS codes (Panel C), and the peer classification of Hoberg and Phillips (2010) according to a product similarity measure (Panel D). Results are again insignificant when we calculate $Equity\ shock^{ACN}(pseudo-peer)$ over firms that are in the same two-digit SIC code as the analyst peer firms and that do not have any common analyst with the firm in question. The only coefficients that are significant in the right direction are obtained when we focus on firms within the same GICS industries. Here we find that there is a negative association between $Equity\ shock^{ACN}(pseudo-peer)$ and *Leverage*, but no relation for leverage changes and equity issuances. We also find a positive and significant coefficient for *Gross Equity* (and weakly significant for the conditional debt-equity choice) when we define pseudo-peers using the Hoberg-Phillips industry peer definition, but not for leverage (levels or changes) or net equity issues (Panel D). To summarize, we obtain very weak evidence for an association between $Equity\ shock^{ACN}(pseudo-peer)$ and firm financial policies. This is in contrast to the strong relation between a firm's financial policies and those of the firms that are

Table 9. Placebo Test with Pseudo Peers

	Δ Leverage (1)	Leverage (2)	Net Equity (3)	Gross Equity (4)	Net Debt (Issuer sample) (5)
Panel A: Three-digit SIC code					
Equity shock ^{ACN} (pseudo-peer)	−0.010 (0.012)	−0.011 (0.014)	−0.063 (0.038)	−0.016 (0.042)	0.032 (0.064)
Observations	40,428	40,428	40,428	40,428	19,606
R ²	0.117	0.769	0.393	0.459	0.615
Panel B: Two-digit SIC code					
Equity shock ^{ACN} (pseudo-peer)	−0.025 (0.019)	−0.030 (0.024)	−0.132 (0.066)**	−0.038 (0.070)	0.131 (0.109)
Observations	40,146	40,146	40,146	40,146	19,484
R ²	0.114	0.77	0.394	0.459	0.614
Panel C: Fama-French industry classification					
Equity shock ^{ACN} (pseudo-peer)	0.033 (0.019)*	0.006 (0.023)	−0.193 (0.066)***	−0.104 (0.069)	0.124 (0.105)
Observations	39,865	39,865	39,865	39,865	19,347
R ²	0.112	0.771	0.397	0.461	0.617
Panel D: GICS industry classification					
Equity shock ^{ACN} (pseudo-peer)	−0.017 (0.015)	−0.053 (0.017)***	−0.039 (0.052)	0.086 (0.054)	−0.045 (0.080)
Observations	40,257	40,257	40,257	40,257	19,583
R ²	0.112	0.769	0.399	0.462	0.616
Panel E: Hoberg-Phillips peers					
Equity shock ^{ACN} (pseudo-peer)	−0.005 (0.012)	−0.007 (0.015)	0.052 (0.045)	0.111 (0.048)**	−0.138 (0.073)*
Observations	34,942	34,942	34,942	34,942	17,233
R ²	0.117	0.767	0.404	0.467	0.62
Own characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Industry characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Pseudo-peer characteristics	Yes	Yes	Yes	Yes	Yes

Notes. The table presents the results of OLS regressions that relate pseudo peer firm idiosyncratic equity shock to firm capital structure. The outcome variables are Δ Leverage (column (1)), Leverage (column (2)), Net Equity (column (3)) and Gross Equity (column (4)). Column (5) isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). The main independent variable is Equity shock^{ACN} (pseudo-peer) that is the simple average equity shock of all firms that are in the same industry as a firm's analyst peers but that do not have any common analysts with the firm in question. We use alternate industry definitions to identify the pseudo-peers in the different panels. All control variables are lagged one period, except the number of firm's connections. In the specification with Δ Leverage as the dependent variable all the control variables are in first difference form, except the number of firm's connections and analyst coverage; and we include year and industry fixed effects. The remaining specifications include firm and year fixed effects. In all the specifications we include as control variables individual firm, industry and pseudo-peer characteristics (similar specification as in Table 5). Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity, we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

in the same industries as the pseudo peers, but that are in the analyst network of firm i . This suggests that our previous results were not simply driven by economic connections between industries, but that analyst networks play a particular role in propagating financial policies across firms.

Finally, in Table IA-7 in the online appendix, we estimate similar regressions as the reduced form models in Table 5 but include only the average equity shock of industry peers to compare the strength of peer effects between industry peers and analyst network peers. The results indicate that the influence of peer firms on issuance choice is substantially larger through analyst coverage networks than through industries, and this is

driven largely through a stronger effect on equity issuance decisions. This suggests that peer effects in equity issuance decisions are especially salient when information flows through analyst coverage networks, consistent with the analysts' focus on covering equities and their role in reducing information asymmetry around security offerings.

4.4.2. Exogenous Variation in Common Analysts. To further address the concern that our results are driven by omitted variables associated with analysts choosing to cover economically connected firms, we exploit brokerage house closures and mergers as a source of exogenous variation in the analyst network. If common

analysts transmit peer effects, then an exogenous reduction in common analysts should weaken the influence of peer firms on an individual firm's decision.

We identify the closure and merger of brokerage houses following Kelly and Ljungqvist (2012) (see appendix A of their paper) and use their methodology to identify the set of analysts (and affected firms) that stopped their coverage as a consequence of the brokerage shock.

We follow the pair model methodology proposed by recent papers that use exogenous shocks such as the death of directors/executives and the brokerage closures to capture how shocks in connection among firms lead to changes in commonalities of corporate policy decisions (Fracassi 2017), to changes in information spillovers of common analysts (Martens and Sextroh 2021), and to changes in stock return comovement (Israelsen 2016).

First, we identify and keep pairs of firms with at least one common analyst that was affected by a brokerage house closure/merger. Specifically, we classify firm-pairs as treated in two ways: (1) pairs of firms that were connected by one analyst who stopped the coverage of at least one of the two firms (i and/or k) as a consequence of the brokerage shock and (2) pairs of firms that were connected by one analyst who stopped following firm i and firm k at the same time, which is a more demanding definition.

Because many brokerage closures/mergers take place over a period of several months, and typically the connection is broken or weakens the year after the merger/closure of the brokerage house, we define the year of the shock in the analyst network as the year after the brokerage event. Moreover, as a consequence of using the lagged peer firms' equity shock, we exclude the year after of the brokerage closure from the analysis. Hence, our sample contains the year before the brokerage event ($t = -1$), the year of the brokerage event ($t = 0$), and years 2 ($t = +2$) and 3 ($t = +3$) after that.

Following Martens and Sextroh (2021) and Israelsen (2016), we perform the following regression to exploit the exogenous changes in common analysts:

$$\begin{aligned} y_{ikt} = & \beta_0 + \beta_1 Post(t+2)_{ikt} \times Peer\ Equity\ shock_{kt-1} \\ & + \beta_2 Post(t+3)_{ikt} \times Peer\ Equity\ shock_{kt-1} \\ & + \beta_3 Post(t+2)_{ikt} + \beta_4 Post(t+3)_{ikt} \\ & + \beta_5 Peer\ Equity\ shock_{kt-1} + \alpha_1 Equity\ shock_{ijt-1}^{IND} \\ & + \alpha_2 Equity\ shock_{ijt-1}^{OWN} + \gamma'_1 X_{kt-1} + \gamma'_2 X_{ijt-1}^{IND} + \gamma'_3 X_{it-1} \\ & + \delta' u_{ik} + \phi' v_t + \epsilon_{ikt}. \end{aligned} \quad (3)$$

Equation (3) is similar to the reduced form regression in Table 5, although each observation is at the firm-pair level (firm i and firm k); $Post(t+2)$ and $Post(t+3)$ are dummy variables that takes the value of one in years 2 and 3 after the brokerage shock (and zero otherwise), respectively. Moreover, we include year (v_t) and firm-pair (u_{ik}) fixed effects, and we control for industry and

firm-specific *Equity shock* and characteristics (we also include firm k 's characteristics, X_{kt-1}).²² If common analysts play a role in facilitating peer effects, we expect the influence of peer firms' *Equity shock* on financial policy decisions of firm i to become weaker after the brokerage closure, given the loss of common analysts. Hence, we expect coefficients β_1 and β_2 to have the opposite sign from β_5 . Consistent with our previous results (Table 5), we expect a negative (positive) sign for β_5 when we consider leverage (equity issuances) as the outcome variable.

Table 10 presents the results. In even columns we perform the regression using the first definition of treated firm-pairs, whereas in odd columns we restrict the sample to the second definition. In columns (1) to (4), the outcome variable is *Leverage* (change and level). We find that the sign of β_1 and β_2 associated with interaction terms $Post(t+2) \times Peer\ Equity\ shock$ and $Post(t+3) \times Peer\ Equity\ shock$ have the opposite sign of β_5 and they are statistically significant. In addition, the results are stronger when using the second definition (more demanding) of treated firm-pairs. Moreover, we find that *Peer Equity shock* has the same sign as in Table 5 in all the columns. For equity issuances and conditional debt-equity choice, shown in columns (5) to (10), we also find that β_1 and β_2 have the opposite sign of β_5 , but the results are only statistically significant for equity issues where a common analyst stopped her coverage of both firms at the same time. In sum, our results suggest that after brokerage closures/mergers and the attendant reduction in common analysts, there is a reduction in the influence of peer firms. This effect is more pronounced when common analysts stopped their coverage of both firms simultaneously. These results suggest that analyst connections play a particular role in facilitating peer effects and that our earlier results are not entirely driven by economic linkages that we do not observe.

4.4.3. Analyst Influence. If analyst networks are important in transmitting corporate policy decisions from one firm to another, then the characteristics of the analyst herself may be important for the strength of these peer effects. More influential analysts should be more effective at transmitting policy-relevant information across firms. Firms are also likely to take the comments of such analysts more seriously. We construct two measures that capture the potential influence of analysts. Specifically, from *Institutional Investor* magazine we collect the names of the top four analysts (first, second, third, and runner-up) for each industry during 1990–2015. We classify an analyst as being influential from the first year the analyst appears in the *Institutional Investor* ranking. We classify brokerage houses that employ three or more influential analysts as *All-star brokerage houses*. These roughly represent about 10% of all brokerage houses in our sample. We differentiate between all-star brokerage houses and

Table 10. Pair Model Regression and Brokerage Closures

	Δ Leverage		Leverage		Net Equity		Gross Equity		Net Debt (Issuer sample)	
	1 (1)	2 (2)	1 (3)	2 (4)	1 (5)	2 (6)	1 (7)	2 (8)	1 (9)	2 (10)
$Post(t+2) \times Peer\ Equity\ shock$	0.007 (0.004)*	0.018 (0.006)***	0.014 (0.003)***	0.027 (0.006)***	−0.025 (0.014)*	−0.071 (0.025)***	−0.019 (0.015)	−0.065 (0.027)**	0.013 (0.020)	0.036 (0.039)
$Post(t+3) \times Peer\ Equity\ shock$	0.019 (0.003)***	0.026 (0.006)***	0.012 (0.004)***	0.016 (0.007)**	−0.022 (0.014)	−0.083 (0.026)***	−0.011 (0.014)	−0.068 (0.026)***	0.014 (0.019)	0.005 (0.036)
$Post(t+2)$	−0.007 (0.002)***	−0.014 (0.003)***	0.006 (0.002)***	0.007 (0.004)*	0.014 (0.007)*	0.011 (0.013)	0.037 (0.008)***	0.035 (0.014)**	−0.017 (0.011)	−0.002 (0.018)
$Post(t+3)$	−0.003 (0.002)*	−0.016 (0.003)***	0.008 (0.003)***	0.008 (0.005)	0.014 (0.008)	0.014 (0.016)	0.045 (0.009)***	0.027 (0.018)	−0.030 (0.012)**	−0.018 (0.023)
$Peer\ Equity\ shock$	−0.008 (0.002)***	−0.015 (0.004)***	−0.005 (0.002)**	−0.007 (0.004)*	0.006 (0.009)	0.029 (0.017)*	0.014 (0.009)	0.030 (0.016)*	−0.005 (0.011)	−0.014 (0.021)
$Equity\ shock^{IND}$	−0.015 (0.004)***	−0.036 (0.008)***	−0.025 (0.003)***	−0.028 (0.008)***	0.085 (0.015)***	0.153 (0.032)***	0.129 (0.015)***	0.162 (0.033)***	−0.226 (0.026)***	−0.228 (0.060)***
$Equity\ shock^{OWN}$	0.007 (0.002)***	0.006 (0.003)	−0.021 (0.002)***	−0.026 (0.003)***	0.059 (0.007)***	0.082 (0.012)***	0.087 (0.007)***	0.105 (0.011)***	−0.090 (0.009)***	−0.113 (0.015)***
Own characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Peer and Industry characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year and firm-pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	29,480	8,816	29,480	8,816	29,480	8,816	29,480	8,816	14,096	4,196
R^2	0.296	0.317	0.859	0.859	0.458	0.492	0.586	0.594	0.702	0.7

The table presents the results of the following pair-model regression:

$$y_{ikt} = \beta_0 + \beta_1 Post(t+2)_{ikt} \times Peer\ Equity\ shock_{kt-1} + \beta_2 Post(t+3)_{ikt} \times Peer\ Equity\ shock_{kt-1} + \beta_3 Post(t+2)_{ikt} + \beta_4 Post(t+3)_{ikt} \\ + \beta_5 Peer\ Equity\ shock_{kt-1} + \alpha_1 Equity\ shock_{ijt-1}^{IND} + \alpha_2 Equity\ shock_{ijt-1}^{OWN} + \gamma_1' X_{kt-1} + \gamma_2' X_{ijt-1}^{IND} + \gamma_3' X_{it-1} + \delta' u_{ik} + \phi' v_i + \epsilon_{ikt}.$$

Notes. The outcome variables are Δ Leverage (columns (1) and (2)), Leverage (columns (3) and (4)), Net Equity (columns (5) and (6)), and Gross Equity (columns (7) and (8)). Columns (9) and (10) isolate the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). The main independent variables are $Peer\ Equity\ shock_{kt-1}$ and $Equity\ shock_{ijt-1}^{IND}$, which measure the lagged averages of the idiosyncratic equity shock for analyst peer firm k and industry peers, respectively. Also, $Equity\ shock_{ijt-1}^{OWN}$ measures the lagged averages of the idiosyncratic equity shock of firm i . All control variables are lagged one period, except the number of firm's connections. In the specification with Δ Leverage as the dependent variable all the control variables are in first difference, except the equity shock and the number of firm's connections and analyst coverage. We classify firm-pairs as treated in two ways: (1) pairs of firms that were connected by one analyst who stopped the coverage of at least one of the two firms (i and/or k) as a consequence of the brokerage shock; and (2) pairs of firms that were connected by one analyst who stopped following firm i and firm k at the same time. $Post(t+2)$ and $Post(t+3)$ are dummy variables that takes the value of one in year two and three after the brokerage shock (and zero otherwise), respectively. In all the specifications we include directed firm-pair and year fixed effects. Standard errors are clustered at directed firm-pair level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity, we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

non-all-star brokerage houses to see if there is any difference in the extent of peer effects within their networks. Next, we differentiate analysts based on their level of experience. Each year, we define analysts to have more (less) experience if they are above (below) sample median in terms of the number of years since they first appear on IBES.

Table 11 examines the impact of all-star brokerage houses (Panel A) and analyst experience (Panel B) on the strength of the analyst network peer effect. In Panel A, we present the results of the reduced form in which we include the weighted average equity shock for the two groups of peers (All-Star and Non All-Star). The All-Star average is calculated over peers that share at least one analyst from an all-star brokerage house, whereas the non-all-star average is calculated over peers connected only by analysts not from all-star brokerage houses. For outcome variables in columns (1) to (4) (first difference

and level of leverage, net and gross equity issuance), we find a larger coefficient on the averages for peers connected through analysts from all-star brokerage houses relative to peers connected through non-all-star brokerage houses, although we find that the coefficients are statistically different only for leverage and net equity issuances. The one exception is for the conditional debt-equity choice in column (5), suggesting that connections through all-star analysts may be more influential for equity issuance decisions than for debt decisions.

Similar, but stronger results are obtained in Panel B where we differentiate analysts based on their experience (more experienced versus less experienced).²³ In all specifications, we find stronger peer effects among firms that are connected through more experienced analysts. All these differences are statistically significant, with the exception of the coefficient for net equity issuances. Interestingly, the peer effect is not statistically different from

Table 11. Cross-Sectional Tests: All-Star Brokerage Houses and Analyst Experience

	Δ Leverage (1)	Leverage (2)	Net Equity (3)	Gross Equity (4)	Net Debt (Issuer sample) (5)
Panel A: All-star brokerage houses (all-star vs. non all-star)					
<i>Equity shock</i> ^{ACN} (All-Star)	−0.018 (0.005)***	−0.023 (0.006)***	0.051 (0.018)***	0.054 (0.019)***	−0.048 (0.028)*
<i>Equity shock</i> ^{ACN} (Non All-Star)	−0.013 (0.004)***	−0.007 (0.004)	0.008 (0.014)	0.034 (0.014)**	−0.066 (0.021)***
(All-Star)-(Non All-Star)	−0.005	−0.016**	0.04**	0.020	0.018
Own characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Peer characteristics (all-star)	Yes	Yes	Yes	Yes	Yes
Peer characteristics (non all-star)	Yes	Yes	Yes	Yes	Yes
Industry characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Observations	41,411	41,411	41,411	41,411	20,150
R ²	0.118	0.768	0.395	0.458	0.614
Panel B: Analyst experience (more experienced vs. less experienced)					
<i>Equity shock</i> ^{ACN} (More experienced)	−0.027 (0.005)***	−0.026 (0.006)***	0.037 (0.018)**	0.064 (0.019)***	−0.082 (0.028)***
<i>Equity shock</i> ^{ACN} (Less experienced)	−0.004 (0.003)*	−0.001 (0.003)	0.011 (0.010)	0.012 (0.011)	−0.028 (0.016)*
(More)-(Less)	−0.022***	−0.024***	0.026	0.052**	−0.054*
Own characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Peer characteristics (more experienced)	Yes	Yes	Yes	Yes	Yes
Peer characteristics (less experienced)	Yes	Yes	Yes	Yes	Yes
Industry characteristics (including equity shock)	Yes	Yes	Yes	Yes	Yes
Observations	41,411	41,411	41,411	41,411	20,150
R ²	0.118	0.768	0.396	0.458	0.613

Notes. The table presents the results of OLS regressions that relate peer firm average financial policy and idiosyncratic equity shock to firm capital structure. The outcome variables are Δ Leverage (column (1)), Leverage (column (2)), Net Equity (column (3)), and Gross Equity (column (4)). Column (5) isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred (*Net Debt (Issuer sample)*). We classify brokerage houses that employ at least three all-star analysts as all-star brokerage houses. We classify analysts with above median experience as being more experienced. In Panel A, we differentiate *Equity shock*^{ACN} across all-star and non-all-star brokerage houses, whereas in Panel B, we differentiate it across more and less experienced analysts. All control variables are lagged one period, except the number of firm's connections. In the specification with Δ Leverage as the dependent variable, all the control variables are in first difference form, except the number of firm's connections and analyst coverage; and we include year and industry fixed effects (three-digit SIC code). The remaining specifications include firm and year fixed effects. Standard errors are clustered at the firm level. All variables are winsorized at the 1st and 99th percentile and are defined in Appendix A. For brevity, we suppress the constant. Standard errors are in parentheses. Statistical significance at the 10%, 5% and 1% levels are denoted by *, ** and ***, respectively.

zero for firms connected through less experienced analysts, but always significant for firms connected through more experienced ones.

4.4.4. Other Robustness Tests. We perform a number of additional robustness tests whose results are presented in the online appendix. We briefly discuss their results here. Recent literature shows that firms with common institutional shareholders tend to follow similar financial policies (Cronqvist and Fahlenbrach 2008). In Table IA-8 of the online appendix, we repeat our tests after including the characteristics of firms that share institutional shareholders with the firm in question and find our results on analyst peers to be robust. Consistent with prior literature we find evidence for peer effects within institutional shareholder networks. We also repeat our tests with four alternate industry definitions: two-digit SIC code, Fama-French industry classification, GICS codes, and Hoberg-Phillips peers. We find our results to be robust across

these industry classifications (Tables IA-9 to IA-11 in the online appendix).

5. Conclusion

Sell-side analysts are important information intermediaries in financial markets. There is growing evidence that they may influence the financial policies of firms that they cover. In this paper, we provide evidence that sell-side analysts are an important mechanism underpinning peer effects in financial policy choices. Building on recent empirical methods from the network effects literature to identify peer effects, we find that exogenous changes to financial policies of firms covered by an analyst, such as leverage and equity issuance, lead other firms covered by the same analyst to make similar changes in policy.

We use an extended Manski-type linear-in-means model and use the idiosyncratic equity shocks of analyst peer firms, as well as the return shocks of indirect peers

(“friends of friends”), as instruments for analyst peer firm financial policies. We show that the network effects that we document are distinct from industry peer effects and that these effects are more pronounced among peers connected by analysts that are more experienced and from more influential brokerage houses. Moreover, the peer effects are weaker after the reduction in common coverage as a consequence of brokerage closures.

An important question that we leave for future research is to establish if the propagation that we document is value enhancing or value destroying. Prior research suggests that such mimicking behavior could be value-enhancing or not, depending on the mechanism driving the behavior. For example, an observational learning channel may allow firms to get closer to optimal capital structure, despite uncertainty about the underlying model (Devenow and Welch 1996). Conversely, mimicking that results from irrational herding may not bring firms closer to optimal policy. Although our empirical approach does not enable us to examine value effects directly, our results show a positive association between peer equity shocks and SEO announcement returns. This suggests that information relevant for equity issuance decisions that is intermediated through analysts may be more likely to be value enhancing. Future research can also explore if there is similar propagation in other corporate policies such as investment and governance provisions such as design of executive compensation. Apart from research analysts, firms are also connected by other channels such as social ties or commonality of board of directors, executives, commercial/investment bankers or other professional advisors, and institutional or active investors. The methodology used in this paper can be fruitfully used to identify peer effects in these other settings.

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Appendix A. Variable Definitions

- *Coverage*: Number of analysts issuing EPS forecasts of firm i in a given year.
- *Equity Shock*: Idiosyncratic returns defined as the difference between realized and expected returns based on the methodology provided by Leary and Roberts (2014).

- *Gross Equity*: Dummy variable that takes the value of one if gross equity issuances *normalized by* book assets at the beginning of the year is greater than a threshold (*Compustat items: sstk/at(t - 1) > 1%, 3%, 5%*).
- *Leverage*: The ratio of the sum of total long-term debt plus total debt in current liabilities scaled by the market value of assets (*Compustat items: (dltt + dlc)/(prcc_f*cshpri + dlc + dltt + pstkl-txditc)*).
- *Log(Sales)*: Natural logarithmic of sales (*Compustat items: log(sale)*).
- *Market to Book*: The ratio of the sum of the total book value of debt plus market value of equity divided by book value of total assets (*Compustat items: (prcc_f*cshpri + dlc + dltt + pstkl-txditc)/at*).
- *Market Value of Assets*: The sum of the market value of equity plus total long-term debt plus current liabilities (*Compustat items: prcc_f*cshpri + dlc + dltt + pstkl-txditc*).
- *Net Debt Issuances*: The sum of the total long-term debt plus total debt in current liabilities for the current fiscal year minus the sum of the total long-term debt plus total debt in current liabilities in the previous fiscal year (*Compustat items: (dltt + dlc - (dltt(t - 1) + dlc(t - 1)))*).
- *Net Debt*: Dummy variable that takes the value of one if net debt issuances *normalized by* book assets at the beginning of the year is greater than 1%. (*Compustat items: (dltt + dlc - (dltt(t - 1) + dlc(t - 1)))/at(t - 1) > 1%*).
- *Net Equity Issuances*: Difference between equity issuances and equity repurchases (*Compustat items: sstk-prstk*).
- *Net Equity*: Dummy variable that takes the value of one if net equity issuances *normalized by* book assets at the beginning of the year is greater than a threshold (*Compustat items: (sstk-prstk)/at(t - 1) > 1%, 3%, 5%*).
- *Profitability*: The ratio of the EBITDA divided by book value of total assets (*Compustat items: oibdp/at*).
- *Stock Return*: Annual return for the firm's stock over the current fiscal year (*Compustat items: ((prcc_f/ajex + dvpsx_f/ajex)/(prcc_f(t - 1)/ajex(t - 1)) - 1)*).
- *Tangibility*: The ratio of the book value of Net Property Plant and Equipment divided by book value of total assets (*Compustat items: ppent/at*).

Endnotes

¹ Examples include Matvos and Ostrovsky (2010); Shue (2013); Leary and Roberts (2014); Kaustia and Rantala (2015); and Fracassi (2017).

² Or perhaps both. For example, a shock to a peer firm's investment opportunities that generates a positive return shock may affect the peer's investment behavior and elicit an equity issuance to fund the investment.

³ See, for example, Frankel et al. (2006); Kadan et al. (2012); Muslu et al. (2014); Chang et al. (2006); and Piotroski and Roulstone (2004).

⁴ Recent research has shown that analysts can influence firm policies either directly by exerting influence (Degeorge et al. 2013, Bradley et al. 2017) or indirectly, if managers alter firm policies to meet analyst forecasts (Hribar et al. 2006, Bhojraj et al. 2009, Gunny 2010). We differ from this earlier work in that we study the information flow from one firm to another in an analyst network.

⁵ We conduct our tests using traditional industry classification measures based on three-digit SIC codes and alternative industry classifications such as the Fama-French industry classifications, the Hoberg-Phillips industry peers (Hoberg and Phillips 2010, 2016),

and S&P GICS codes to allow for the possibility that alternative industry grouping better capture economic commonalities across firms. Our results on the analysts' peer effect are robust across these various industry definitions.

⁶ In additional robustness test, we repeat our estimates after excluding all indirect peers in the same industry as firm i .

⁷ Although the policies of peer firms may be public knowledge, analysts may still play an important role in communicating the nuances and value relevance of the policy to other firms, enabling management to better assess the suitability of the policy for the firm.

⁸ Kaustia and Rantala (2015, 2021) also examine peer effects within the context of analyst coverage networks. However, their focus is on stock split decisions, and they use analyst networks to identify groups of related firms rather than study the role of analysts in transmitting information from one firm to another.

⁹ As we explain later, in reality, our sample starts in 1987 but we require five years of data to calculate the rolling betas (we use a window of 60 months). Hence, to estimate the idiosyncratic equity shock of the year 1992, the first rolling windows is 1987–1991. In 1993, we have all the observations for the lagged equity shock and control variables and the contemporaneous outcome variables.

¹⁰ In all the regressions, we use the 1% threshold for the gross and net equity (debt) issuances to define the indicator variable. We explicitly identify the cases in which we use a different threshold.

¹¹ We also create an alternative measure of industry average outcomes that includes only firms that are in the same industry as firm i but are not in the same analyst network as firm i . In other words, we exclude the set of firms that overlap across the analyst coverage network and industry of firm i .

¹² Leary and Roberts (2014) show evidence that this strategy produces idiosyncratic return estimates that are uncorrelated, both serially and cross-sectionally, within networks.

¹³ In each year, we calculate monthly peer returns using the firm's analyst network in that year. To calculate $\bar{r}_i^{ACN}$, we require that a firm has at least one peer firm with valid returns during the time period in which we estimate the loadings.

¹⁴ One potential concern could be that analysts may not pass information about a financial policy change at firm i on to other firms in their network if they know the change was precipitated by a shock that was idiosyncratic to firm i . However, first, although the analyst observes the policy change itself, the analyst may not be aware at the time whether it was associated with a purely idiosyncratic shock. Second, Leary and Roberts (2014) show evidence that corporate managers do respond to financing decisions associated with idiosyncratic shocks at peer firms, suggesting it is plausible that analysts would also.

¹⁵ Consistent with the simple average idiosyncratic shock of indirect peers, for the friend-of-friends approach we run the 2SLS with the simple average of peer firm outcome and controls variables. However, the results remain unchanged if we use peer-weighted averages.

¹⁶ On average, only 7% of indirect peers are in the same industry (three-digit SIC code) as firm i .

¹⁷ We also include the own firm's equity shock ($Equity\ shock^{OWN}$) as an additional firm characteristic.

¹⁸ In subsequent analyses, we continue to utilize the conditional debt issuance specification from column (6) in place of the specification in column (5).

¹⁹ In these regressions, when we control for industry average financial policy, we focus on firms that do not have common analysts because the policy of firms with common analyst (and in the same industry) is the endogenous variable and is instrumented by the indirect peer average.

²⁰ In columns (1)–(3), we repeat the results in Table 8, but we restrict our sample to firms with also non-missing indirect peers' equity shocks when we impose the restriction of at least two (and four) direct peers. Thus, our results are comparable across columns.

²¹ Because there are no analyst connections for this latter group, we use the simple average of their idiosyncratic shocks rather than weighting by the number of common analysts.

²² Following Martens and Sextroh (2021), standard errors are clustered at the directional firm-pair level.

²³ We repeat the process and create two weighted average equity shocks for the two peer groups: $Equity\ shock^{ACN} (More\ experienced)$ and $Equity\ shock^{ACN} (Less\ experienced)$.

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